
Evaluating Deep Features for Predicting Video Quality

Subtitle: Deep features for modeling human visual perception

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I certify that except where due acknowledgement has been given, the work presented in this thesis is that of the author alone; the work has not been submitted previously, in whole or in part, to qualify for any other academic award; and the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program.

Kaan Egemen Sen
Lugano, 15 January 2024

It is strange that only
extraordinary men make the
discoveries, which later appear so
easy and simple

Georg C. Lichtenberg

Abstract

The growing demand for high-quality video content necessitates advancements in how video quality is assessed. While perceptual loss has proven effective for image quality evaluation, its application to video content remains underexplored. This thesis introduces a novel framework for perceptual video quality assessment, inspired by perceptual loss concepts originally developed for images. Despite its success in image processing, the extension of perceptual loss to video has not been thoroughly investigated, leaving a gap in understanding how such ideas translate to dynamic visual content.

This thesis bridges this gap by systematically investigating the suitability of deep learning architectures for perceptual video quality assessment. Inspired by perceptual loss for images, this work extends the concept to video by examining FlowNet and ViNet—two models chosen for their distinct capabilities. FlowNet captures motion coherence through optical flow estimation, while ViNet predicts attention-driven saliency, focusing on regions that attract human attention. Together, these models address the complex spatial and temporal dynamics of video content. By leveraging intermediate feature maps, this framework evaluates perceptual differences to quantify the impact of distortions such as compression artifacts and noise on video quality.

A key focus of this research is the analysis of frame lengths, demonstrating how processing sequences of frames enhances the alignment between computational metrics and human perception. Results reveal that deep features from FlowNet and ViNet significantly improve the predictive power of quality metrics, achieving a strong correlation with human ratings. This framework provides valuable insights into the design of perceptual video quality assessment methods, advancing beyond pixel-level or structural comparisons.

By extending perceptual loss to dynamic video content, this thesis contributes a novel approach to video analysis, with applications in adaptive streaming,

compression optimization, and real-time video enhancement. The findings not only validate the effectiveness of using deep features for perceptual assessment but also highlight opportunities for further innovation in creating immersive and perceptually optimized video experiences.

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Chapter 1

Introduction

Videos appear to present smooth motion and vivid detail, mimicking real-life experiences. However, they are constructed from separate frames made of pixels, and the perceived quality relies heavily on the level of detail these pixels can capture. While advancements in technology have elevated the demand for high-quality video content, meeting this demand requires substantial bandwidth, computational resources, and storage capacity. This growing need underscores the importance of balancing video quality with system efficiency.

Reducing video quality can make systems lighter, but determining the optimal balance is critical. Video quality metrics play a pivotal role here, offering quantifiable insights into how much quality can be reduced without significantly affecting user perception. These metrics help evaluate the trade-offs between system resources and perceived quality, enabling efficient content delivery.

Beyond resource optimization, quality metrics are also indispensable in evaluating image and video processing methods. Whether for compression, enhancement, or generation, these metrics provide objective benchmarks for assessing performance. Moreover, they are integral to training procedures, where robust metrics enable better optimization of AI models, enhancing their ability to learn from and replicate human perception.

By developing and employing accurate video quality metrics, we not only address issues like blurriness, noise, and compression artifacts but also ensure that static quality and motion fidelity are maintained. These metrics bridge the gap between computational evaluations and human perception, facilitating advancements in video analysis, enhancement, and AI-driven solutions.

1.1 Context

Assessing video quality comes with specific challenges because videos are not like still images. Videos consist of a series of frames that must look good individually, but they also need to work well together. The movement and changes in scenes matter a lot, so the assessment of video quality needs to consider how these elements interact.

Many traditional methods of evaluating video quality rely on checking pixels or segmenting images. However, these methods often miss what is important for how viewers actually experience videos. The way we perceive videos is impacted by factors such as motion and what stands out in the scene. Traditional metrics may not account for these important aspects.

Recently, deep learning models have changed how we think about image quality. Tools like Learned Perceptual Image Patch Similarity (LPIPS) [26] use deep features to match assessments with how people perceive images. Although this method has been effective for images, applying it to videos is more complicated. Videos need tools that can understand not just the details in each frame, but also how frames relate to each other over time.

In video quality estimation, factors like optical flow and saliency play a big role. Optical flow refers to the motion between frames in a video, while saliency looks at parts of the image that catch our attention. Bringing these factors into the assessment improves the understanding of video quality. To this end, we need better models that can pick out important features from video.

Our goal is to address these challenges by advancing from traditional image analysis to techniques leveraging deep features specifically tailored for videos. In image quality evaluation, perceptual loss (which is originally designed for images) has demonstrated significant utility by identifying differences between original and distorted content using deep feature representations from pre-trained classification models. These models are trained to perform tasks that humans excel at, such as object recognition, and inherently encode key image features like texture, structure, and focus that align with human perception. This makes perceptual loss particularly effective in capturing details that pixel-based metrics often overlook.

However, applying perceptual loss to videos introduces unique questions. Unlike static images, videos have an additional temporal dimension, making it unclear what constitutes the most relevant features for evaluating video quality. Should the focus be on optical flow to capture motion coherence, saliency to identify attention-driven regions, or some other factor entirely? Addressing these questions is critical for developing metrics that accurately represent

human perception in dynamic video content.

Building on this idea, we apply the concept of perceptual loss to videos, where motion and time are just as critical as spatial detail. To achieve this, we utilize models like FlowNet, which captures motion coherence through optical flow, and ViNet, which identifies attention-grabbing areas in video frames. While ViNet is often motivated by its ability to highlight important regions, we view it as more than just a saliency model, it is a network trained for a task that humans inherently excel at: focusing attention on key features within a scene. This alignment with human perception makes it particularly effective for encoding perceptual traits that are crucial for evaluating video quality. By integrating these deep features into a unified framework, we can comprehensively assess video quality, accounting for both spatial and temporal dimensions in a manner that resonates with human visual processing.

1.2 Contribution

We aim to explore the relationship between model performance and its alignment with human perception in the context of video quality estimation. Specifically, our research investigates the following questions:

- Can deep representations from models trained on video tasks like optical flow estimation (FlowNet) and saliency prediction (ViNet) effectively predict the perceptual quality of videos?
- How do the number of frames used for feature extraction influence the correlation between computational metrics and human visual perception?
- Do better-performing models (e.g., those with lower endpoint error or higher saliency prediction accuracy) exhibit stronger alignment with human-perceived quality?
- What aspects of spatio-temporal features, such as motion coherence and attention-driven saliency, are critical for improving perceptual correlation scores?
- How do distortions such as noise, compression artifacts, or motion inconsistencies affect the ability of deep features to encode perceptual differences?

- Can optimizing the combination of deep features from multiple models (e.g., FlowNet and ViNet) improve the prediction of perceptual quality beyond individual model performance?

To achieve our goals, we made several important contributions:

- **Deep Feature Utilization for Perceptual Loss:** We used advanced video models like FlowNet and ViNet to get feature maps. These features show details about motion and important areas in the video. This helps us calculate perceptual loss in great detail.
- **Framework for Video Quality Estimation:** We created a new way to estimate video quality by connecting feature-based error metrics with scores given by humans. This framework helps link model predictions to how people actually perceive video quality. It offers a strong method for video analysis.
- **Evaluation of Temporal Length:** We carefully looked at how using sequences of input frames can improve correlation scores. Our results show that analyzing more frames together leads to better accuracy in perceptual metrics. This highlights the importance of context when assessing video quality.

These contributions improve the field of perceptual video quality estimation. They provide a way to combine computer techniques with how people view videos. By focusing on motion, important areas, and the flow of time, our work sets the stage for better video analysis in many areas, such as streaming services and real-time video enhancement.

1.3 Structure of the thesis

In this work, we will present the following

- **Chapter 1: Introduction**

The introduction offers a summary of the research. It discusses important ideas about video quality assessment and highlights the difficulties that video content can bring. The need for perceptual metrics that match how people see things is emphasized. Additionally, the introduction explains what the thesis contributes and prepares readers for the chapters that follow.

- **Chapter 2: Background**

Background provides important information to understand the themes in this thesis. First, it looks at the concept of image processing. It gives general information about Convolutional Neural Networks. Next, it introduces key concepts like optical flow, saliency prediction, and perceptual loss. These elements help create a clearer picture of video quality and its measurement.

- **Chapter 3: Related Works**

This chapter reviews existing literature on video quality estimation and deep learning methods for perceptual metrics. It explores the use of LPIPS [26] and [23] Why are deep representations good perceptual quality features? papers. Additionally, it examines optical flow estimation and saliency detection. The chapter highlights how these techniques are used in video processing and discusses their limitations.

- **Chapter 4: Methodology**

Methodology explains the method used for the thesis. It describes the models that were applied, such as FlowNet and ViNet. These models help in pulling deep features from videos. A new framework for judging video quality is also introduced. This framework uses perceptual loss to enhance the analysis. Additionally, the chapter discusses how to assess the effects of temporal length, which refers to the number of input frames, and how this is added to the final evaluation metric.

- **Chapter 5: Experimentation and Results**

This chapter covers the experiments carried out to test the proposed method. It provides information about the datasets used, how the evaluation metrics were chosen, and the setup for the experiments. The results show how well the new video quality estimation framework performs compared to older methods. Special focus is given to the benefits of including optical flow and saliency in the analysis.

- **Chapter 7: Discussion**

This chapter discusses the findings from the experiments, including insights into how motion and saliency affect video quality perception. It also reflects on the strengths and limitations of the proposed approach and suggests potential improvements and future research directions.

- **Chapter 8: Conclusions**

The final chapter brings together the key points from the thesis. We

emphasize how important it is to include perceptual loss, motion, and saliency when estimating video quality. This research could lead to better video enhancement techniques. It also has implications for compression methods and streaming services. As video content continues to grow, understanding quality perception will help improve viewer experiences. Future research might explore even more ways to enhance video quality, making it enjoyable for all users.

Chapter 2

Background

2.1 Image Processing

Before engaging in video processing, it's crucial to grasp the fundamental concepts of image processing. Image processing refers to the technique of optimizing a captured image for specific purposes through the application of various computer algorithms and visual methods on digital media. Not every image is guaranteed to be usable. With the advancements in image processing technology, any image can now be tailored for its intended application. The captured image is translated into a function by computer algorithms. The image is viewed in two dimensions, where each color and every point is considered as components of the function, allowing algorithms to manipulate this function effectively.

Image processing is basically analysed in five stages:

- Acquisition: The digital image is acquired with a digital camera.
- Pre-Processing: Before using the digital image, it is necessary to go through some pre-processing. These are image enhancement, image restoration, image compression. Acquiring and pre-processing the image is called low-level image processing.
- Segmentation: The subheadings of this process are; segmentation, segmentation, segmentation. This part is the more difficult part of image processing. The aim of this part is to detect and separate the different properties of objects and areas in the image. With the detection of object and area boundaries in the image, it produces raw - unprocessed - information on the shape. If the aim is to find the boundaries, edges, corners

of the objects in the image, it is detected by segmentation. At the same time, features such as the area and colour of the object in the image are obtained by regional segmentation method. Both methods are used to solve problems such as character or pattern recognition.

- Inference: It is the highlighting of the raw information obtained, different desired features and details.
- Interpretation: In this last part, different objects and areas in the image processed with some artificial intelligence algorithms are classified and labelled.

2.2 Convolutional Neural Networks

To begin with, when discussing artificial neural networks as a whole, these models are inspired by how the human brain operates. The goal of this system is to carry out the learning process, understand the acquired knowledge, and independently make decisions based on this understanding. Essentially, this architecture is made up of three layers.

- Input Layer
- Hidden Layer
- Output Layer

In the input layer, the initial observations are sent to the system. The quantity of nodes in this layer corresponds to the number of attributes needed to effectively represent the observation. In the input layer, values can be duplicated and passed on to multiple layers. No processing occurs during the value transfer stage. The hidden layer takes the values from the input layer, multiplies them by specific coefficients, and applies transformation operations to those values. This layer may contain several nodes. By implementing specific threshold conditions on these nodes, a number of values equal to the outputs can be generated. Following the hidden layer, there is the output layer. Based on the generated value, the system provides a prediction.

Artificial neural networks were initially introduced by [16]. At that time, the limitations of computing power constrained the understanding of the models' capabilities. In the 2000s, a substantial boost in computing power resulted in

increased success and interest in artificial neural networks. Despite this, the human brain contains approximately 100 billion neurons, which serves as a foundational reference for the model. Current systems cannot fully replicate this complex structure. However, these models continue to progress alongside advancements in technology over time.

2.2.1 CNN Layers

Convolutional neural network consists of several layers, some of which are as follows:

- Convolutional Layer
- MaxPooling
- Flattening

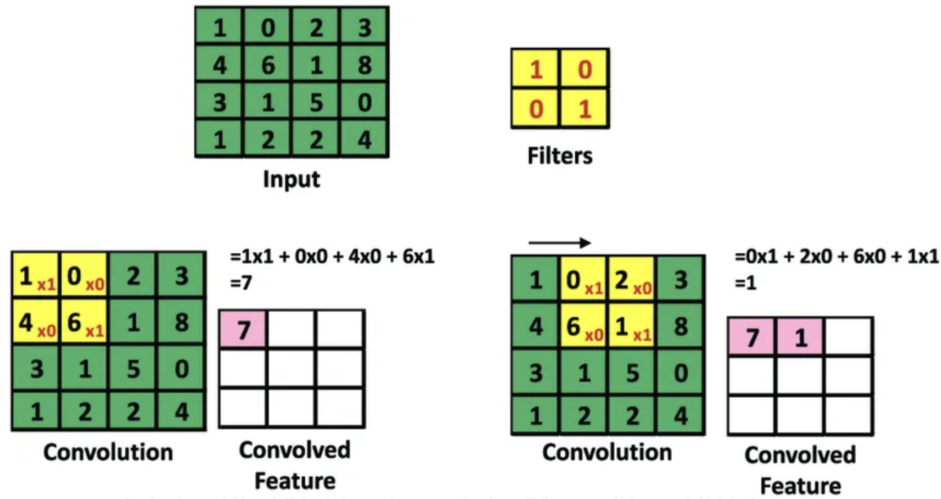


Figure 2.1. Basic Convolutional Layer taken from [17]

To begin with, the **Convolutional Layer** processes the input image by applying a filter. The resulting values from this filtering create what is known as the feature map. This process can be observed step by step in 2.1. We utilize a 4x4 matrix as our input, while the chosen filter measures 2x2 in size. (In practice, filters for real images typically come in sizes such as 3x3 or 5x5. For instance, prominent models like AlexNet utilize 7x7 filters.) In this scenario, the filter is moved across the entire input matrix, and the feature matrix is generated by multiplying the filter matrix with each 2x2 submatrix in the input.

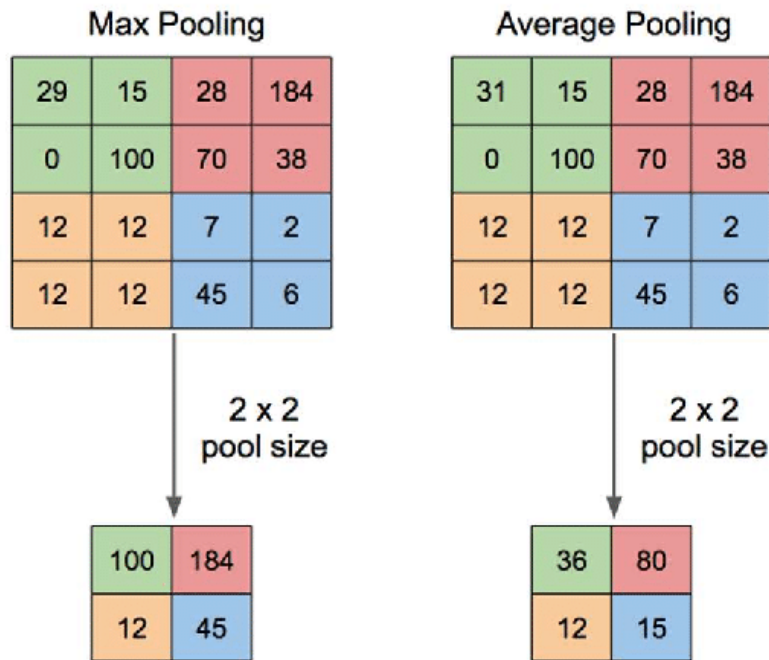


Figure 2.2. Pooling Example taken from [19]

Generally, the next layer after the Convolution layer is the **Pooling layer**. The pooling layer is typically applied to the generated attribute matrices. As illustrated in Figure 4, a 4x4 attribute matrix is presented. When a 2x2 max pooling is applied here, the resulting 2x2 matrix on the left is produced. Within the attribute matrix, segments of the specified pooling dimensions (2x2 for our example) are formed, and the maximum value from each segment is selected. In the initial matrix shown in orange, this maximum value is 12, while it is 100 for the matrix in green, 45 for the blue area, and 184 for the red area. Although max pooling is the most commonly utilized technique, there are alternative methods such as mean pooling and min pooling. In the mean pooling method depicted on the right side of the figure, the average value is calculated, whereas in the min pooling method, the lowest value is selected.

At last, we come to our **Flattening** process.

The preceding layers can be utilized multiple times throughout the model. Once these operations are completed, the resulting matrix needs to be flattened to be compatible with the Fully Connected layer. Therefore, the flattening process, which serves as an illustration of an entire Convolutional Neural Network (see Figure 2.3), is executed. Following this step, the inputs for the Fully Con-

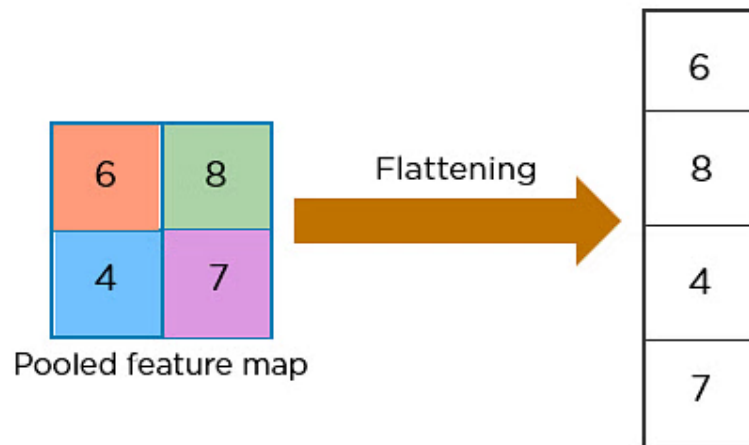


Figure 2.3. Flattening taken from [3]

nected Layer are set up.

2.3 Transformation Functions

Transformation functions are fundamental in computational systems, providing the means to modify and reshape data for effective processing. Unlike simple preprocessing steps, these functions often involve mathematical operations that transform inputs to meet specific needs in various applications, such as neural networks or data analytics workflows.

Transformation functions can introduce non-linearity, scale values, or change data distributions, enabling systems to better handle complex real-world scenarios. Without these functions, raw data might remain limited in usability, reducing the efficiency of models and algorithms. Some of the most notable transformation functions include Softmax, Sigmoid, Arctangent (Arctan), and Linear transformations. These mathematical tools are indispensable in machine learning and artificial intelligence contexts.

2.3.1 Softmax Function

The Softmax function is a transformation that converts logits (raw prediction values) into probabilities. It ensures that the output values sum to 1, making it particularly useful in classification problems involving multiple classes. Mathe-

matically, the Softmax function is expressed as:

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

where x_i represents the input vector, and n is the number of classes. By emphasizing the largest logits while suppressing smaller ones, Softmax 2.4 is essential for probabilistic interpretations in neural networks.

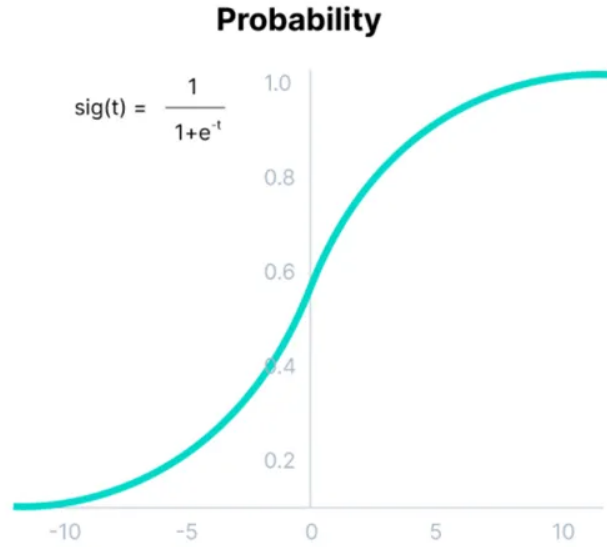


Figure 2.4. Softmax Function taken from [12]

2.3.2 Sigmoid Function

The Sigmoid function 2.5 maps input values to a range between 0 and 1, making it suitable for binary classification tasks. It is defined as:

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

The smooth S-shaped curve of the Sigmoid function helps squash extreme values, but its tendency to cause vanishing gradients can limit its applicability in deep networks.

2.3.3 Arctangent (Arctan) Function

The Arctangent function, or arctan, is a non-linear transformation that maps input values to the range $-\pi/2$ to $\pi/2$. Unlike the Tanh and Sigmoid func-

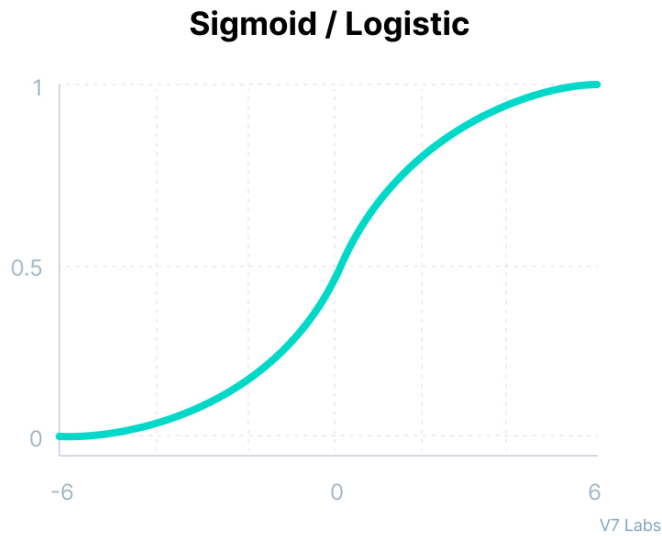


Figure 2.5. Sigmoid Function taken from [12]

tions, which operate on exponential terms, the Arctan function is defined using a trigonometric relationship, making it computationally efficient for certain applications. The formula is:

$$\text{Arctan}(x) = \tan^{-1}(x)$$

This function is particularly useful for applications requiring bounded yet symmetric output ranges. The Arctan function 2.6 is less prone to saturation at extreme values compared to Sigmoid, which can aid in optimization stability. Its continuous and smooth gradient properties make it suitable for various transformation tasks in machine learning and signal processing.

2.3.4 Exponential Function

The Exponential function is a continuous, non-linear transformation that maps input values to a range from 0 to ∞ . It is widely used in various fields, such as machine learning, signal processing, and probability theory. The formula is:

$$\text{Exponential}(x) = e^x$$

The Exponential function grows rapidly for positive values of x and approaches zero as x becomes more negative. Its properties, such as smoothness and continuous growth, make it useful in tasks that require rapid escalation or

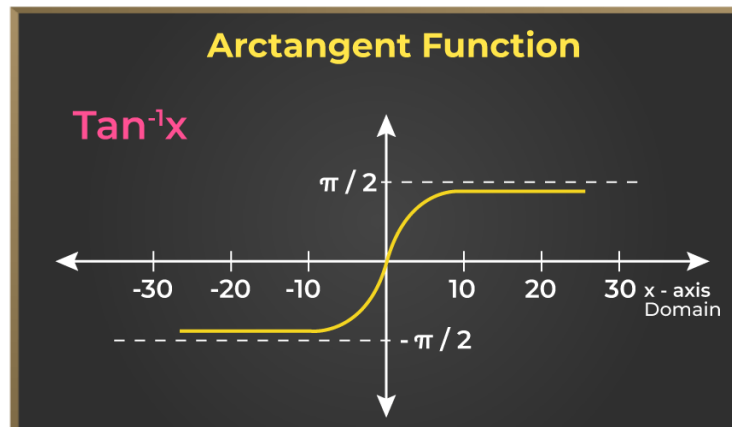


Figure 2.6. Arctan Function taken from [7]

modeling of phenomena such as population growth, financial growth, or activation in neural networks. The Exponential function 2.7 is particularly known for its ability to avoid saturation at extreme values, unlike functions like Sigmoid or Tanh.

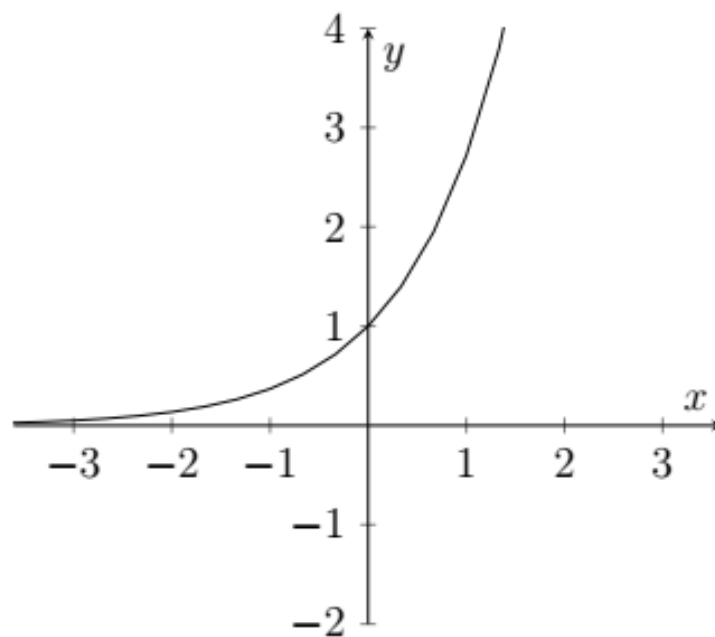


Figure 2.7. Exponential Function taken from [8]

2.3.5 Linear Transformation

Linear transformations apply a simple linear mapping to data, maintaining proportional relationships between input and output. These transformations are mathematically expressed as:

$$y = ax + b$$

where a is the scaling factor, and b is the bias term. While linear transformations do not introduce non-linearity, they are foundational for operations like scaling and feature combination.

By employing these transformation functions, models gain the capability to process data effectively, capturing both linear and non-linear patterns in datasets. Their appropriate use is critical in designing systems that align closely with real-world complexities and requirements.

2.4 Optical Flow

Optical flow is the modeling of the movement of objects between two consecutive frames caused by the movement of the object or camera. In other words, it is the process of distinguishing the moving object when compared to two consecutive frames.

Optical flow works with the following two assumptions:

- The pixel intensities of an object do not change between consecutive frames.
- Neighbouring pixels have similar motion.

The Lukas-Kanade method of the optical flow algorithm assumes that the pixel under investigation is essentially stationary in a local neighbourhood. It also solves the basic optical flow equations for all pixels in that neighbourhood by least squares.

The Lucas-Kanade method takes a 3x3 kernel around the point. In other words, the determined point and its 9 neighbouring points have the same motion. While small motions can be solved with this kernel, pyramids are used for large motions. When we move up in the pyramid, small motions are removed and large motions turn into small motions. Thus, by applying Lucas-Kanade to frames, optical flow is obtained.

An example of optical flow can be seen in the figure 2.8

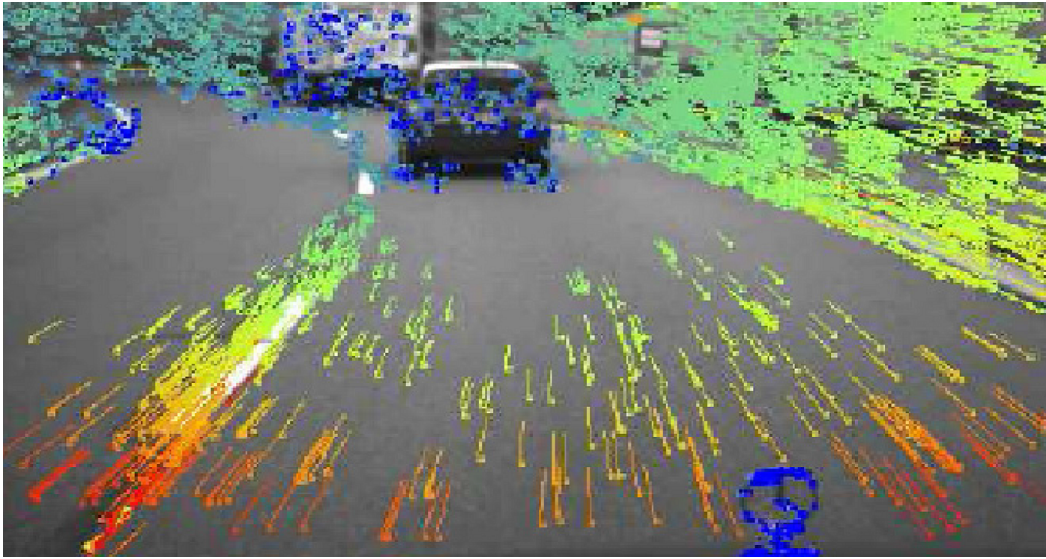


Figure 2.8. Optical Flow taken from [21]

2.5 Video Saliency

Video analysis often measures saliency by looking at the differences between two frames that come right after each other. This technique helps to spot important changes and movements in a video. By finding key areas in one frame and then comparing them with the next, it's easier to catch significant actions.

For instance, think about a video from a security camera monitoring cars in a parking lot. The system looks at the differences from one frame to the next to measure saliency. It first identifies the important areas in the current frame. Then, it compares this with the following frame to see if anything has changed or moved.

If a car leaves a parking spot, this change shows up in the differences between the frames. The saliency map highlights this moving area, helping the system understand what is happening. This approach makes it easier to detect motion and sudden changes since the system focuses only on the significant differences. Another example to this can be seen in the figure 2.9

In summary, saliency maps, created by analyzing the differences between consecutive frames, help to quickly find important movements and events in the video. This method improves video analysis and makes sure that crucial information is easy to see.

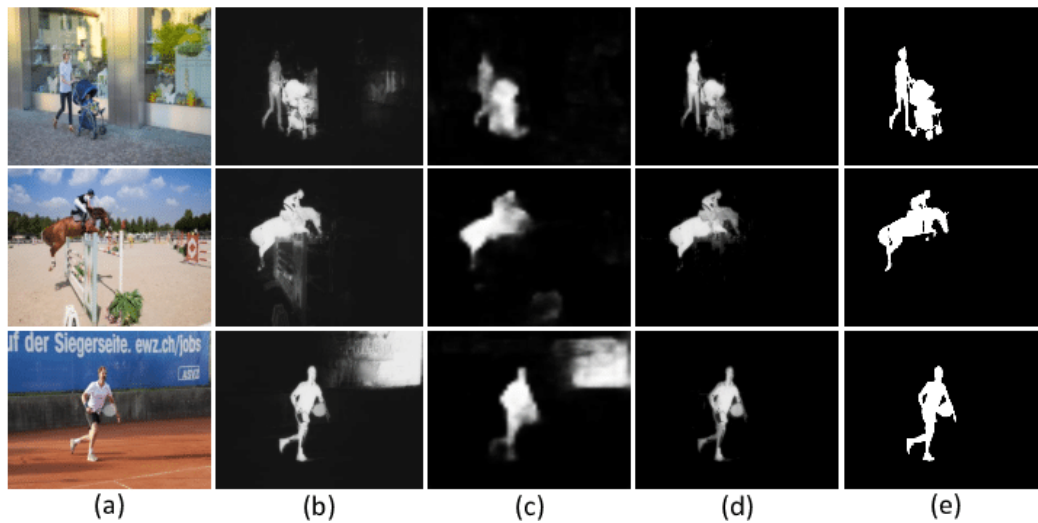


Figure 2.9. Example of Saliency map taken from [22]

2.6 Perception

Perception is a key topic in psychology and cognitive science. It involves how we receive and interpret sensory information from the world around us. This process begins with physical stimulation of our senses. For example, light enters our eyes, and sound waves reach our ears.

However, perception is not just about receiving these signals. Our experiences, attention, memory, and expectations all play a role in how we perceive things. There are two main ways that we process information: bottom-up processing and top-down processing.

Bottom-up processing takes simple details and builds them into a complete understanding. For instance, when we look at an object, we first see colors and shapes before recognizing it. Top-down processing uses what we already know to influence how we interpret what we sense. For instance, if we see something familiar, our past experiences shape our understanding of it.

At the end of the 19th century, experimental psychology began to grow. This led to new ideas about how we understand perception. Different methods helped shape this field. Psychophysics looks at how different physical qualities affect what we perceive. For example, it studies how changes in sound or brightness impact our senses. Sensory neuroscience explores how the brain processes these perceptions. This field examines the brain's functions and how they relate to what we sense.

Another approach is computational analysis, which uses computers to un-

derstand how perception works. In addition, philosophy raises important questions about perception. It asks whether qualities such as sound and smell exist outside our minds or if they are purely subjective experiences.

While we often see our senses as simple tools, research into illusions shows a different story. The brain actively interprets sensory information, creating experiences that sometimes happen without our awareness. There is ongoing debate about whether our perception works as a way to test ideas actively or if getting accurate sensory information makes that testing unnecessary. This question remains an important topic in the study of perception.

The brain helps us make sense of the world around us. It does this even when the information from our senses isn't perfect. Our brains have different areas that handle various types of sensory information. These areas are set up in a way that allows them to work together.

One clear example is how taste and smell work together. When we eat, the way food tastes can change based on how it smells. This connection is important for how we perceive flavors. So, our brains combine these senses to give us a full experience of what we eat.

Perceptual Loss Definition

The perceptual loss, ℓ_{percep} , is computed as the Euclidean distance between feature representations of the original and distorted images. Let $\phi_j(x)$ represent the activations of the j -th layer of a pre-trained network ϕ for an image x . The perceptual loss is defined as:

$$\ell_{\text{percep}}(\hat{y}, y) = \sum_j \frac{1}{C_j H_j W_j} \sum_{c=1}^{C_j} \sum_{h=1}^{H_j} \sum_{w=1}^{W_j} (\phi_j(\hat{y})_{chw} - \phi_j(y)_{chw})^2, \quad (2.1)$$

where C_j , H_j , and W_j denote the dimensions of the feature map at layer j . This loss function captures perceptual differences that are not evident at the pixel level, thus aligning better with human perception.

2.7 Metrics

Spearman Correlation

Spearman's rank correlation coefficient, represented as ρ , is a non-parametric statistic that measures the strength and direction of the monotonic relationship

between two variables. It evaluates how accurately the connection between two variables can be modelled by a monotonic function, indicating that as one variable increases, the other is likely to increase (or decrease) in a consistent way, although not strictly in a linear fashion.

Formula: The formula for Spearman's correlation ([25]) is:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

- d_i is the difference between the ranks of the corresponding variables,
- n is the number of data points.

Interpretation:

- A value of $\rho = 1$ indicates perfect positive rank correlation,
- $\rho = -1$ indicates perfect negative rank correlation,
- $\rho = 0$ suggests no monotonic relationship between the variables.

Spearman correlation is often used in situations where the data is ordinal (rank-based), not necessarily continuous or linear.

Correlation Coefficient

The correlation coefficient is a statistical measure that describes the degree to which two variables move in relation to each other. It is most commonly represented by the Pearson correlation coefficient for continuous data.

Formula: The Pearson correlation coefficient, r , is given by:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Where:

- x_i and y_i are the data points of the two variables,
- $\bar{x}$ and $\bar{y}$ are the means of the variables x and y , respectively.

Interpretation:

- $r = 1$ indicates perfect positive correlation,
- $r = -1$ indicates perfect negative correlation,
- $r = 0$ indicates no linear relationship.

The correlation coefficient quantifies how strongly two variables are related, specifically in a linear manner. A high positive value suggests that the variables increase together, while a negative value suggests that as one variable increases, the other decreases.

Similarity

In the context of data science, similarity generally refers to how alike two objects, vectors, or sets are. It is used across multiple domains like clustering, recommendation systems, and information retrieval. Similarity is typically measured using metrics like:

- **Cosine Similarity:** Measures the cosine of the angle between two vectors, which can range from -1 (completely opposite) to 1 (exactly the same), with 0 indicating no similarity.

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

Where $\mathbf{A}$ and $\mathbf{B}$ are vectors, and $\cdot$ denotes the dot product.

- **Euclidean Similarity:** Measures the straight-line distance between two points in space. The smaller the distance, the higher the similarity.

Similarity is widely applied in clustering algorithms and recommendation systems, where the goal is to group or suggest items that are most alike.

EPE (End-Point Error)

EPE is a common metric used in optical flow and image registration tasks to quantify the accuracy of a predicted displacement field (or flow) compared to the ground truth. It measures the Euclidean distance between corresponding points in the predicted and true flow fields.

Formula:

$$\text{EPE} = \sqrt{(x_{\text{pred}} - x_{\text{gt}})^2 + (y_{\text{pred}} - y_{\text{gt}})^2}$$

Where:

- $(x_{\text{pred}}, y_{\text{pred}})$ are the predicted flow coordinates,
- $(x_{\text{gt}}, y_{\text{gt}})$ are the ground truth flow coordinates.

Interpretation: The EPE measures the average error in the prediction of optical flow or the alignment of two images. Smaller values indicate more accurate predictions, and larger values indicate poor accuracy. It is particularly useful for evaluating how well a model's predicted motion or displacement matches the true motion in a video or image sequence.

Optimization Methods

- Powell Method is iteratively minimizing function through a set of directions. It does not require a gradient of the objective function, making it suitable for non-smooth or noisy problems
- L-BFGS-B uses an approximation of the inverse Hessian matrix to guide optimization, storing only a limited number of vectors to save memory (hence "limited memory")
- TNC is a truncated Newton method that uses the conjugate gradient technique to approximate the solution to the Newton equations. It is a gradient-based optimization method.
- SLSQP solves problems with equality and inequality constraints using a quadratic programming subproblem at each iteration.
- Nelder-Mead evaluates function values at the simplex vertices and performs operations like reflection, expansion, contraction, and shrinking.

Chapter 3

Literature Review

3.1 Perceptual Losses for Real-times Style Transfer and Super-Resolution

To understand image transformation better, the work by Johnson et al. in their paper "Perceptual Losses for Real-Time Style Transfer and Super-Resolution" [11] stands as a significant milestone. This paper is a major milestone in the field. It tackles some of the key limitations of traditional per-pixel loss functions, which are often used to train convolutional neural networks (CNNs) for image transformation tasks. Their method combines the high-quality outputs of optimization-based approaches with the speed of feed-forward networks, making it suitable for real-time applications. This work has had a significant impact on recent research and continues to influence the development of new techniques in image processing and computer vision.

Feature Reconstruction Loss

The feature reconstruction loss aims to ensure that the output image has similar high-level feature representations as the target image. Let $\phi_j(x)$ denote the activations of the j -th layer of the pre-trained network ϕ when processing the image x . The feature reconstruction loss is defined as the squared Euclidean distance between the feature representations of the output image $\hat{y}$ and the target image y :

$$\ell_{\text{feat}}^{\phi,j}(\hat{y}, y) = \frac{1}{C_j H_j W_j} \|\phi_j(\hat{y}) - \phi_j(y)\|_2^2, \quad (3.1)$$

where C_j , H_j , and W_j are the dimensions of the feature map at layer j . This

loss encourages the output image to be perceptually similar to the target image by matching their high-level features.

Style Reconstruction Loss

In addition to preserving content, the style reconstruction loss ensures that the output image matches the style of the target image. This is achieved by comparing the Gram matrices of the feature maps, which capture the correlations between different feature channels and thus encode the style information. The style reconstruction loss for layer j is defined as:

$$G_j^\phi(x)_{c,c'} = \frac{1}{C_j H_j W_j} \sum_{h=1}^{H_j} \sum_{w=1}^{W_j} \phi_j(x)_{h,w,c} \phi_j(x)_{h,w,c'} \quad (3.2)$$

where $G_j(x)$ denotes the Gram matrix computed from the activations $\phi_j(x)$. By minimizing this loss, the method ensures that the output image replicates the texture and style patterns of the target image.

3.2 The Unreasonable Effectiveness of Deep Features as a Perceptual Metric

[26] presents a detailed study of deep network features as a perceptual metric for image similarity, addressing a major challenge in computer vision: aligning algorithmic assessments of image similarity with human perception. Traditional metrics such as PSNR and SSIM often fail to perform well enough to capture the nuances of human visual perception, leading to mismatches between algorithmic results and human judgments. To circumvent this problem, Zhang and colleagues extract deep features from pre-trained networks and propose a method that can leverage these features to compensate for the performance shortcomings of PSNR and SSIM. Their work demonstrates the power of deep visual representations, showing that these features can capture perceptual similarity even in self-supervised and unsupervised learning contexts. Furthermore, the introduction of the BAPPS dataset and a comprehensive assessment framework significantly enriches the field and provides valuable tools for future research.

3.3 Why Are Deep Representations Good Perceptual Quality Features?

[23] undertake a deep investigation into the effectiveness of intermediate feature maps from pre-trained convolutional neural networks (CNNs) in enhancing perceptual quality. Their research addresses a crucial question in computer vision: why do these deep features when used in loss functions, perform better than traditional perceptual metrics such as Structural Similarity Index Measure (SSIM) and Peak Signal-to-Noise Ratio (PSNR)?

3.3.1 Motivation and Background

Traditional image similarity metrics like PSNR and SSIM are widely used but often fall short in capturing human perceptual quality. These metrics primarily focus on low-level pixel-wise differences and basic structural similarities, which do not always align with human visual perception. In contrast, deep features derived from CNNs have been shown to significantly improve perceptual quality for various image processing tasks. For instance, Johnson et al. [11] demonstrated that the distance between feature representations in pre-trained CNNs could serve as an effective perceptual loss function, leading to notable enhancements in image quality. Subsequent studies, including those by Zhang et al. [26] and Ledig et al. [13], have adopted and validated this approach, confirming the superiority of deep features in perceptual quality tasks.

Despite these advancements, there remains a gap in understanding the underlying reasons why deep features perform so well. Tariq et al. aim to bridge this gap by analyzing the characteristics of deep features and their correlation with fundamental human visual perception attributes, specifically contrast sensitivity and orientation selectivity. These attributes are crucial as they are core aspects of the human visual system (HVS) that influence our perception of image quality.

Frequency Selectivity

To measure frequency selectivity, the authors generate sinusoidal gratings of various spatial frequencies and pass them through pre-trained CNNs like VGG-16. By analyzing the activations of intermediate layers, they quantify the sensitivity of each channel to different spatial frequencies. This analysis reveals

which channels are most responsive to frequencies that are critical for human perception, as determined by the Contrast Sensitivity Function (CSF):

$$\mu_1(k, m) = \sum_f CSF(f) \cdot \left| \frac{\partial}{\partial f} a_m^k(I)_f \right|,$$

where k is the index for the convolution layer, m is the feature map index in each convolution layer, $CSF(\cdot)$ is the contrast sensitivity function, a_m^k is the mean activation of the feature map and f is the spatial frequency in cycles per degree. μ_1 quantifies the average frequency sensitivity of a CNN channel weighted by the CSF over different spatial frequencies. Channels with higher μ_1 values should deliver better perceptual features according to the authors' hypothesis.

Orientation Selectivity

Similarly, to assess orientation selectivity, the authors create sinusoidal gratings oriented at various angles. The activations of intermediate layers in response to these gratings are analyzed to determine the orientation selectivity of each channel. This metric is essential for detecting structural distortions, as the human visual system is highly sensitive to specific orientations and edge structures like :

$$\hat{a}_m^k = \max a_m^k(I_\theta)$$

based on $\hat{a}_m^k$ defining orientation selectivity score μ_2 as:

$$\mu_2(k, m) = \sum_{\theta} (a_m^k(I_\theta) - \hat{a}_m^k)^2$$

Just-Noticeable Difference (JND) Tests

In JND tests, subjects are asked to identify the minimum detectable difference between pairs of images. The results show that features with higher frequency and orientation selectivity enable more accurate detection of subtle perceptual differences, highlighting their effectiveness in enhancing image quality.

3.3.2 Implications of Findings

The findings of Tariq et al. have significant implications for various image-processing tasks. The authors provide a framework for selecting and utilizing deep features to optimize perceptual quality by demonstrating the correlation between deep feature selectivity and human visual perception attributes.

Perceptual Efficacy Metric

One of the key contributions of this work is the introduction of the Perceptual Efficacy (PE) metric. This metric combines frequency and orientation selectivity scores to rank the effectiveness of feature channels in optimizing perceptual quality. The PE metric enables the selection of a subset of features that improve image reconstruction quality for tasks like single-image super-resolution, thereby reducing the redundancy often present in large feature sets.

$$PE = (\Phi_m^k) = \frac{\mu_1(k, m) \cdot \mu_2(k, m)}{\sum_m \mu_1(k, m) \cdot \sum_m \mu_2(k, m)} \quad (3.3)$$

Applications in Image Processing

The insights gained from this study can be applied to a wide range of image processing applications, including super-resolution, denoising, inpainting, and style transfer. By using deep feature-based perceptual metrics, researchers and practitioners can better assess and enhance the perceptual quality of their algorithms, leading to more visually appealing results.

Real-Time Applications

The applicability of deep feature-based metrics in real-time image processing tasks is particularly noteworthy. Traditional metrics like PSNR and SSIM are computationally efficient but inadequate for capturing perceptual quality. Deep feature-based metrics, while more computationally intensive, offer a much better perceptual alignment, making them suitable for real-time applications where perceptual quality is paramount.

3.4 Video compression dataset and benchmark of learning-based video-quality metrics

This study introduces a new dataset and a method to benchmark learning-based video quality metrics. It focuses on the quality of video compression. About 2,500 video streams were compressed using different standards like AVC, HEVC, AV1, VP9, and VVC. Researchers collected subjective scores for these videos through crowdsourced matching benchmarks. The main goal was to assess how well the metrics perform, considering different issues from newer compression standards.

The findings indicate that next-generation metrics, which use machine learning and neural networks without full reference, show a strong correlation. This is similar to the results from full-reference metrics. The study highlights that some full-reference metrics, particularly VMAF, work best for assessing compression quality. They show the highest correlation with people's quality ratings. A new benchmarking method is also developed for future use. This method includes both obvious and hidden parts of the dataset to prevent over-learning.

This study serves as a valuable resource for researchers and industry professionals who aim to advance video processing and quality measurement.

3.5 FlowNet: Learning Optical Flow with Convolutional Networks

3.5.1 FlowNet Approach

FlowNet introduces a novel approach to optical flow estimation using convolutional networks [6]. Unlike traditional methods, FlowNet directly learns optical flow from data, leveraging the power of deep learning to capture complex motion patterns.

Learning-based Methods

The key innovation in FlowNet is its ability to learn optical flow estimation from a large dataset of synthetic images with known flow fields. This learning-based approach enables the network to generalize to real-world scenarios, providing robust performance across diverse scenes.

The synthetic dataset used for training FlowNet, known as FlyingChairs, consists of images generated from 3D models of chairs in various configurations

and motion patterns. This dataset provides a wide range of motion scenarios, helping the network learn to handle different types of motion effectively.

Basic Formulation

The fundamental formulation introduced authors considers only a single comparison of two patches. The 'correlation' of two patches centered at x_1 in the first map and x_2 in the second map is then defined as:

$$c(x_1, x_2) = \sum_{o \in [-k, k] \times [-k, k]} (f_1(x_1 + o), f_2(x_2 + o)) \quad (3.4)$$

for a square patch of size $K := 2k+1$. It is identical to one step of convolution in neural networks, but instead of convolving data with a filter, it convolves data with other data. For this reason, it has no trainable weights. Computing $c(x_1, x_2)$ involves $c \cdot K^2$ multiplications.

Supervised Learning Framework

The supervised learning framework used in FlowNet involves training the network on a synthetic dataset with known optical flow. The loss function, typically based on the endpoint error (EPE), guides the network to produce accurate flow estimates that closely match the ground truth.

The architecture of FlowNet includes encoder and decoder networks with skip connections that help in preserving spatial information. The encoder network gradually reduces the spatial dimensions while increasing the feature length, capturing high-level motion features. The decoder network, on the other hand, upsamples the feature maps to the original resolution, producing dense flow estimates.

3.5.2 Comparison with Traditional Methods

FlowNet's performance is evaluated against traditional variational methods and other deep learning approaches. The results demonstrate that FlowNet achieves competitive accuracy while offering significant speed advantages, making it suitable for real-time applications. FlowNet model can be seen in the figure 3.1 below :

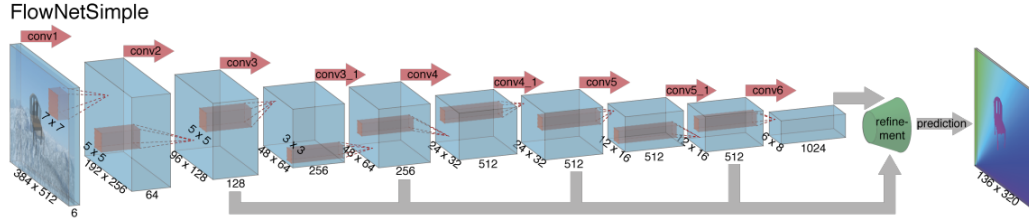


Figure 3.1. Flownet Model Architecture taken from [6]

Improvements and Innovations

Several enhancements have been introduced to variational methods over the years. Notable improvements include advanced regularization techniques and the integration of robust data terms that handle large displacements effectively [4]. Furthermore, the combination of variational approaches with combinatorial matching techniques has yielded significant performance gains, as demonstrated by works such as [24].

One significant advancement is the introduction of non-local regularization methods, which consider the global structure of the image rather than just local smoothness. This approach helps preserve edges and fine details in the flow field, leading to more accurate results in complex scenes.

3.5.3 Deep Matching and DeepFlow

The work by Weinzaepfel et al. [24], known as DeepMatching and DeepFlow, is closely related to the FlowNet approach. This method aggregates feature information from fine to coarse scales using sparse convolutions and max-pooling. However, unlike FlowNet, DeepFlow does not incorporate learning; all parameters are manually set.

3.6 FovVideoVDP

In the "FovVideoVDP" [15] paper, the authors propose a novel visual difference metric that possesses several key features. This metric is calibrated in physical units, ensuring its relevance to real-world perception. It also models temporal dynamics, accounting for how humans perceive visual changes over time. By accommodating foveated viewing, it considers how we view images with our central and peripheral vision. Importantly, the metric demonstrates robustness across diverse types of content and outperforms traditional metrics that rely on

hand-crafted features or statistical indicators. Furthermore, it can be efficiently computed on a GPU, making it one of the fastest metrics of its complexity. The need for this new metric arises in graphics applications, particularly in AR/VR rendering, where foveated viewing and spatio-temporal artifacts play a crucial role. Looking ahead, the authors plan to use this metric as a differentiable loss function to optimize both traditional and machine learning techniques. However, it currently does not model certain aspects of vision, such as color, glare, inter-channel masking, and eye motion, which they aim to address in future work.

3.7 Spatio-Velocity Contrast Sensitivity Function (CSF)

The aim of this research is to investigate how the velocity of eye movements influences contrast sensitivity, particularly in the context of motion blur on LCD televisions. By conducting a series of experiments that incorporate eye-tracking, we seek to enhance the existing 2D spatio-velocity contrast sensitivity function (CSF) model. This study will determine whether eye velocity affects sensitivity and assess the model's applicability to more complex visual stimuli. The experiments will utilize Gabor patterns and a disembodied edge to populate and validate the CSF model, ultimately optimizing it for our experimental conditions.

3.8 Perceptual Visual Quality Metrics

The study of perceptual visual quality metrics has gained significant attention in recent years due to its critical role in various applications, including image and video compression, transmission, and enhancement. The paper "Perceptual Visual Quality Metrics: A Survey" by Weisi Lin and C.-C. Jay Kuo, published in the *Journal of Visual Communication and Image Representation*, provides an extensive review of existing metrics designed to assess visual quality from a human-centric perspective. This section discusses the evolution, methodologies, and applications of perceptual visual quality metrics, drawing heavily from the survey by Lin and Kuo in [14]

The evolution can be categorized into three main phases:

1. **Pixel-Based Metrics:** MSE and PSNR, despite their simplicity, serve as the foundation for understanding the need for more perceptual approaches.

2. **Structural Similarity Metrics:** The introduction of the Structural Similarity Index (SSIM) marked a significant improvement by considering changes in structural information, which is more aligned with human perception.
3. **HVS-Based Models:** Advanced metrics such as Visual Information Fidelity (VIF) and the Visual Signal-to-Noise Ratio (VSNR) incorporate comprehensive models of the HVS, addressing factors like contrast sensitivity and visual masking.

3.8.1 Methodologies in Perceptual Quality Assessment

Perceptual visual quality metrics are designed to simulate various aspects of the HVS to predict perceived visual quality accurately. The methodologies employed in these metrics can be broadly categorized as follows:

- **Frequency-Domain Analysis:** Techniques such as the Discrete Cosine Transform (DCT) and Discrete Wavelet Transform (DWT) decompose images into frequency components. Metrics like VIF utilize these transforms to assess visual quality by comparing the significant frequencies perceived by the HVS.
- **Contrast Sensitivity Function (CSF):** The CSF models the varying sensitivity of the HVS to different spatial frequencies. Metrics incorporating CSF can better predict perceptual quality by weighting errors according to their visibility to the human eye.
- **Visual Masking:** This phenomenon occurs when the presence of certain visual patterns makes other patterns less noticeable. Metrics such as SSIM and VIF consider visual masking to provide more accurate quality assessments, especially in textured regions.
- **Natural Scene Statistics (NSS):** Metrics based on NSS leverage the statistical properties of natural images. By comparing the statistical deviations in distorted images to those in natural scenes, metrics like the Natural Image Quality Evaluator (NIQE) provide perceptual quality predictions without needing a reference image.

3.8.2 Comparative Analysis of Metrics

- **SSIM**: Widely adopted due to its simplicity and effectiveness in capturing structural information. However, it may not fully account for all HVS characteristics.
- **VIF**: Offers a comprehensive approach by modeling the HVS and considering information fidelity. It is computationally intensive, which can be a limitation for real-time applications.
- **MS-SSIM**: An extension of SSIM that evaluates images at multiple scales, improving robustness in different viewing conditions. It balances performance and computational complexity effectively.
- **PSNR-HVS**: Enhances traditional PSNR by incorporating HVS-based weighting. It provides a good balance between simplicity and perceptual accuracy.
- **NIQE**: Unique in its no-reference approach, relying solely on NSS. While versatile, it may not perform as well in highly controlled scenarios where a reference image is available.

3.9 ViNet: Pushing the limits of Visual Modality for Audio-Visual Saliency Prediction

The ViNet paper [10] introduces a fully convolutional encoder-decoder architecture designed for visual saliency prediction. Unlike traditional audio-visual saliency models, ViNet achieves state-of-the-art performance using only visual input, making it a significant contribution to the field of saliency prediction.

3.9.1 Visual Saliency Prediction

Saliency prediction focuses on modeling human visual attention, aiming to identify regions in a visual scene that are likely to attract attention. Previous approaches in video saliency prediction rely on Long Short-Term Memory (LSTM) networks to model temporal dependencies. However, these methods often treat spatial decoding and temporal aggregation separately, limiting their effectiveness.

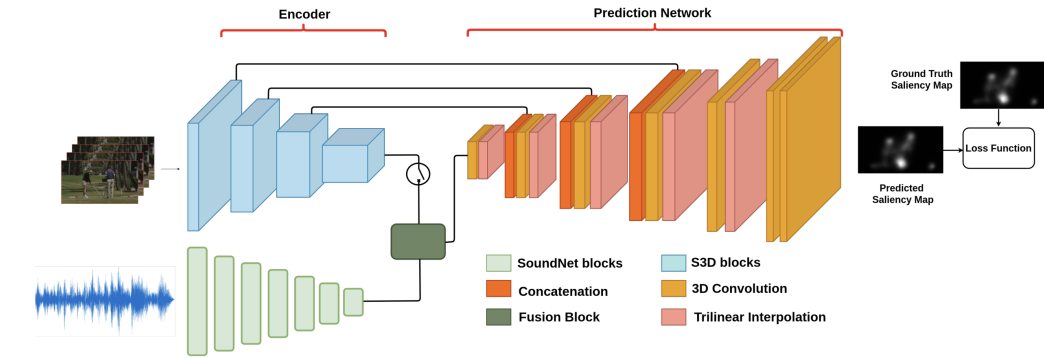


Figure 3.2. ViNet model Architecture taken from [10]

3.9.2 Architecture

ViNet introduces a novel approach by using a fully convolutional network that directly predicts saliency maps. The architecture incorporates features from a pre-trained action recognition network, which captures multi-scale hierarchical features. The encoder-decoder architecture processes a sequence of frames and generates saliency maps through trilinear interpolation and 3D convolutions, making it conceptually simple and computationally efficient and Architecture of the model can be seen below 3.2:

3.9.3 Performance and Results

ViNet significantly outperforms existing models on both visual-only and audio-visual datasets, despite not using any audio input. Notably, it surpasses human performance on the AVE dataset in metrics such as Pearson Correlation Coefficient (CC), Similarity (SIM), and Area Under the Curve (AUC). These results demonstrate the effectiveness of focusing on visual features for saliency prediction, challenging the traditional belief in the necessity of audio input for this task.

Chapter 4

Methodology

In our approach, we want to correlate error calculation in intermediate feature maps of video frames and their mean opinion score. But to be able to start working with video frames we decided first to trial image frames and their distorted versions that are also used in [23], which we mentioned in the Related Works section, we compared Mean Opinion scores of images with error scores obtained from comparisons of intermediate layer features extracted from the VGG model with the original image and its distorted version. At the end of these processes, we were able to achieve the following correlation ratio

After successfully completing this step, it was important to determine the right model and dataset for the next stage. For this purpose, we decided to focus on two types of models that are most suitable and ideal for videos: video flow computation and video saliency computation models.

The model we used for video flow computation was able to help us achieve our goals due to its simple architecture and being one of the first flow models. This model was the FlowNet model 3.5 mentioned earlier [9]. FlowNet processes video frames in pairs calculates the optical flow between these frames, and produces estimated flow frames.

ViNet [10], our chosen model 3.9 for computing video saliency, is favoured for its ability to handle complex visual patterns and its performance to accurately predict regions of interest in dynamic scenes. ViNet builds on current visual saliency methods by effectively utilising temporal coherence and spatial awareness, which are critical for accurately identifying saliency in video data. The model's architecture, designed to capture attention-oriented features in both the temporal and spatial dimensions, is perfectly aligned with our need to analyse how distortions affect viewer focus and attention in high-motion real-

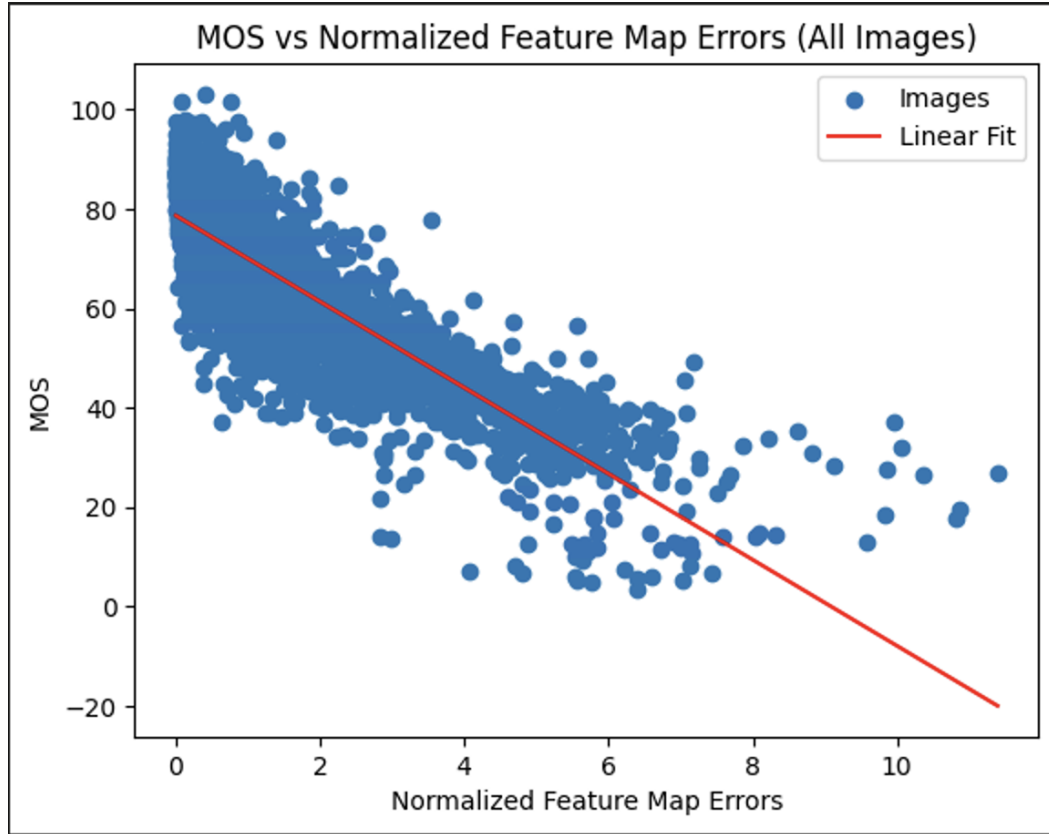


Figure 4.1. Spearman correlation of normalized feature maps with MOS: -0.8536713426564616

world video. These features make ViNet an ideal model for analysing saliency and distortion sensitivity in a variety of video content.

Using this model, we wanted to access not only the final output data but also the feature maps produced in the intermediate layers. To achieve this, we implemented a ‘hook’ mechanism in the various layers of the model. The process of extracting the feature maps consists of several steps: first, we set up the storage structure, then we add forward hooks, perform a forward pass of the model, and finally, we remove the hooks. This allows us to examine and analyze the intermediate feature maps produced at each layer. The mentioned function can be seen in the algorithm 1 below :

After obtaining the feature maps from clean and corrupted video frames, the next step is to compare these feature maps and measure the difference between them. This comparison allows us to understand how much the neural

Algorithm 1 Extract feature maps from specified layers of FlowNet model

```

model: FlowNet model
input_pair: Pair of input frames
layers: List of layer names for feature map extraction
feature_maps: Dictionary of extracted feature maps
feature_maps ← {} {Initialize dictionary to store feature maps}
hooks ← [ ] {List to store forward hooks}
for each layer in layers do
    layer_module ← getattr(model, layer)
    if layer_module exists then
        hook ← layer_module.register_forward_hook(hook_fn)
        hooks.append(hook)
    end if
end for
Disable gradient computation
Perform forward pass: model(input_pair)
for hook in hooks do
    hook.remove() {Remove hooks after the forward pass}
end for
return feature_maps {Return the extracted feature maps}
  
```

network's feature representations are affected by the distortions in the input frames. Thus, we can assess the robustness of the model and how sensitive it is to input changes. Such distortions are particularly important for compression artifacts, noise, or degradation in frame quality, which are common in areas such as video processing.

To make this comparison, we perform pixel-wise error calculations between feature maps extracted from clean video (let us call them 'original feature maps') and feature maps extracted from distorted video ('distorted feature maps'). For each layer, we take the feature maps from both clean and distorted frames and calculate the difference between them. This difference is calculated on an item-

by-item basis between pixels in each feature map, i.e. we subtract the value of each pixel in the distorted map from the corresponding pixel in the original map. This creates a difference tensor that shows how much each pixel is degraded compared to the clean map.

We then square the differences to eliminate negative values and emphasize larger deviations. This method makes large errors more prominent. Mathematically, this process is expressed by the equation

$$\text{Error}(i, j) = (F_{\text{original}}(i, j) - F_{\text{distorted}}(i, j))^2$$

In this equation, $F_{\text{original}}(i, j)$ is the pixel value at position (i, j) in the original feature map, and $F_{\text{distorted}}(i, j)$ is the corresponding pixel value in the distorted map. The error for each pixel is calculated by squaring the difference between these two values. By performing this process for all pixels in the map, we obtain a matrix of squared differences that shows the total error magnitude in the feature map.

The dimensions of the feature maps may differ between layers due to neural network subsampling or pooling operations. Given these differences, we need to normalise the errors to make the comparison between layers fair. For this purpose, we sum the squared differences in each feature map and divide this sum by the total number of pixels in the feature map. This method allows us to calculate the **normalized error**, which is a single value representing the mean squared difference for the whole layer. The process is expressed by the equation:

$$\text{Normalised Error} = \frac{1}{N} \sum_{i,j} (F_{\text{original}}(i, j) - F_{\text{distorted}}(i, j))^2$$

Here, N represents the total number of pixels in the feature map (i.e. $N = \text{height} \times \text{width} \times \text{channels}$).

This normalised error metric allows us to more fairly compare the distortion effect between feature maps of different sizes. After calculating this error for each layer, we obtain a corresponding error value for each layer. These values help us to understand how the distortion in the input affects different layers. For example, layers closer to the network can often show larger errors if the model is sensitive to low-level features. In contrast, deeper layers, because they form more abstract representations, may be more resilient to distortions in the input and exhibit lower errors. Pseudo codes of error computation for the Flow model

can be seen in the below pseudo code 2 :

Algorithm 2 Compute errors between original and distorted feature maps

```

original_feature_maps: Dictionary of original feature maps
distorted_feature_maps: Dictionary of distorted feature maps
layer_errors: List of normalized errors for each layer
layer_errors  $\leftarrow$  [ ] {Initialize list to store layer errors}
for each layer_name in original_feature_maps.keys() do
    original_map  $\leftarrow$  original_feature_maps[layer_name]
    distorted_map  $\leftarrow$  distorted_feature_maps[layer_name]
    if original_map.shape  $\neq$  distorted_map.shape then
        distorted_map  $\leftarrow$  interpolate(distorted_map.unsqueeze(0), size = original_map.shape[1:], mode = 'bilinear', align_corners = False).squeeze(0)
    end if
    difference  $\leftarrow$  original_map - distorted_map
    squared_error  $\leftarrow \sum(\text{difference}^2)$ 
    normalized_error  $\leftarrow$  squared_error / numel(original_map)
    layer_errors.append(normalized_error.item())
end for
return layer_errors {Return list of normalized errors}

```

This code computes errors between feature maps extracted from clean and distorted video frames in the FlowNet model. Initially, dictionaries of original and distorted feature maps are used, with each entry corresponding to a specific layer. For each layer, the feature maps are compared by calculating pixel-wise differences between the maps, then squaring these differences to create a matrix of squared errors. When feature map dimensions differ, an interpolation step resizes the distorted map, ensuring accurate comparison. Finally, the sum of squared differences for each feature map is normalized by dividing it by the total pixel count, yielding a single error value per layer. This normalized error indicates the extent of distortion impact, enabling layer-wise resilience analysis across the model.

In addition to FlowNet, we also implemented Vinet as a second model fo-

cused on video saliency computation, providing complementary insights on video frame saliency and attention patterns 3.

For the Vinet model, we iterate over each layer to compute the normalized error between the original and distorted feature maps. First, each feature map is retrieved from the dictionaries, and if the shapes differ, we resize the distorted map to match the original. This adjustment is handled according to the dimensions of the feature map: 5-dimensional maps (e.g., batch, channel, time, height, width) are resized using 'trilinear' interpolation, while 4-dimensional maps (batch, channel, height, width) use 'bilinear'. After resizing, we compute the pixel-wise squared difference and sum it to obtain the total squared error. To normalize, we divide the squared error by the number of elements in each map, adjusting for its dimensionality. This normalized error provides a comparable metric across layers, stored in **layer_errors** for each layer.

After documenting the layer-wise errors for every video, we evaluate the degree of alignment between the scores derived from the feature map errors and the ground truth scores. This evaluation is essential for understanding the precision of our model's predictions and the extent to which these predictions correspond with the perceived quality (ground truth scores) in real-world scenarios. This phase enables us to gain a deeper insight into the model's performance.

4.0.1 Calculation and Comparison of Unweighted Scores

In the initial phase of the analysis, we calculate **unweighted scores**, which represent the average errors of the feature maps across all layers for each video. These scores are derived as a straightforward average of the errors for every video, without any specific weighting applied. We subsequently compare these unweighted scores against the ground truth scores to evaluate their level of agreement. To visualize this correspondence, we generate a scatter plot that displays the unweighted scores along the x-axis and the ground truth scores along the y-axis.

Moreover, we include a linear regression line in the scatterplot that accurately represents the data using the function '`np.polyfit()`'. This line graphically illustrates the relationship between the acquired scores and the actual scores, aiding in our understanding of their correlation. When the data points are closely packed around the line, it signifies a strong correspondence between the two score sets.

4.0.2 Weighted Scores Calculation via Optimization

While the unweighted scores provide a foundational assessment of the relationship between the feature maps and the ground truths, we further refine this connection by weighting the feature contributions of different layers. To enhance this process, we employ various **non-linear transformations** that adjust the weighted scores better to reflect the underlying relationships with the ground truths.

Non-Linear Transformations

We explore several non-linear functions to transform the scores after applying weights. These transformations allow the model better to capture complex relationships between features and the ground truth. The transformations used are as follows:

1. **Sigmoid Transformation:** The sigmoid function normalizes the weighted scores, bounding them between 0 and 1. It is defined as:

$$S(x) = \frac{1}{1 + e^{-k(x-m)}}$$

where k is the steepness and m is the midpoint of the sigmoid curve.

2. **Arc Tangent Transformation:** This transformation uses the arctangent function to capture non-linear relationships. It is expressed as:

$$A(x) = \frac{2}{\pi} \arctan(kx)$$

where k controls the steepness of the transformation.

3. **Softmax Transformation:** The softmax function scales the weighted scores into probabilities that sum to 1. It is given by:

$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

4. **Linear Transformation:** As a baseline, this transformation retains the direct relationship between the weighted scores and outputs:

$$L(x) = x$$

5. **Exponential Transformation:** The exponential transformation amplifies differences in scores by exponentially scaling the values:

$$E(x) = e^x$$

Optimization Strategy

The optimization process aims to determine the weights for feature layers and parameters for non-linear transformations. The goal is to **maximize the Spearman correlation** between the transformed scores (computed from the weighted feature maps) and the ground truth scores. Since optimization algorithms typically minimize functions, we reformulate the objective to minimize the **negative Spearman correlation**:

$$\text{Objective Function: } -\text{Spearman}(T(\mathbf{W} \cdot \mathbf{X}), \text{Ground Truth Scores})$$

Here, T represents the non-linear transformation (e.g., sigmoid, arc tangent, softmax), $\mathbf{W}$ is the weight vector, and $\mathbf{X}$ denotes the feature map matrix.

To ensure stability and meaningful results, constraints are applied to weights and transformation parameters during optimization.

The optimization is performed using the **Powell method**, which is particularly effective for unconstrained optimization problems. Other methods, such as SLSQP and L-BFGS-B, are also tested to ensure consistency and robustness.

4.0.3 Results and Visualizations

After optimization, we compute the weighted and transformed scores for each video. As with the unweighted scores, we visualize the relationship between these scores and the ground truth. Scatter plots are overlaid with regression lines, showing the degree to which the transformed scores align with the ground truth rankings. By leveraging these non-linear transformations, we demonstrate improved alignment between predictions and ground truth scores, reflecting the strength of this approach in capturing complex relationships.

This comprehensive strategy enhances the model's capacity to assess motion and saliency effectively, bridging the gap between the raw feature outputs and the nuanced understanding required for real-world scenarios.

Algorithm 3 Compute Errors Between Original and Distorted Feature Maps with Shape Handling

```

original_feature_maps: Dictionary of original feature maps
distorted_feature_maps: Dictionary of distorted feature maps
layer_errors: List of normalized errors for each layer
layer_errors  $\leftarrow$  [ ] {Initialize list to store layer errors}
for each layer_name in original_feature_maps.keys() do
    original_map  $\leftarrow$  original_feature_maps[layer_name]
    distorted_map  $\leftarrow$  distorted_feature_maps[layer_name]
    if original_map.shape  $\neq$  distorted_map.shape then
        if original_map.shape is 5-dimensional then
            distorted_map  $\leftarrow$  interpolate(distorted_map,
size=original_map.shape[2:], mode='trilinear', align_corners=False)
        else if original_map.shape is 4-dimensional then
            distorted_map  $\leftarrow$  interpolate(distorted_map,
size=original_map.shape[2:], mode='bilinear', align_corners=False)
        else
            raise ValueError {Unexpected shape, handle error}
        end if
    end if
    difference  $\leftarrow$  original_map - distorted_map
    squared_error  $\leftarrow \sum(\text{difference}^2)$ 
    if original_map.shape is 5-dimensional then
         $B, C, T, H, W \leftarrow$  dimensions of original_map
        normalized_error  $\leftarrow$  squared_error / ( $B \times C \times T \times H \times W$ )
    else if original_map.shape is 4-dimensional then
         $B, C, H, W \leftarrow$  dimensions of original_map
        normalized_error  $\leftarrow$  squared_error / ( $B \times C \times H \times W$ )
    end if
    layer_errors.append(normalized_error.item())
end for
return layer_errors {Return list of normalized errors}

```

Chapter 5

Experimental Setup

5.1 Data

In this section, we provide detailed information about the dataset we used in our experiments. We chose the Video Quality Metrics Benchmark dataset, which allows us to test our model extensively with videos of different content and degradation levels. The dataset consists of 17 original reference video clips, and each reference video is reconstructed at 28 different degradation levels. This versatile data structure allows us to compare with various quality control models and measure the accuracy of quality prediction with high confidence. Example snippets of the dataset can be seen in 5.1

One of the major factors in our choice to use this dataset is that there is already a competition organised with this dataset. As a result of this competition, the achievements of various quality control models have been reported in detail, providing data that we can use as a reference to evaluate the performance of our model. This comparison provides an important opportunity for us to analyse the performance of our model against other models and to identify areas for improvement.

The rich variety of content offered by the Video Quality Metrics Benchmark dataset is another factor that increases the applicability of our experiments to real-world scenarios. The dataset contains many different visual elements such as human faces, animals, natural landscapes, artificial structures, and video game scenes. This diversity allows our model to be tested on a versatile dataset and makes it suitable for a wide spectrum of visual stimuli, preventing our predictions from being limited to a specific type of content. This shows that our

model is suitable for a more general use and can produce effective results in different video content types in various sectors.

Each video in the dataset is approximately 30 seconds long, resulting in an average of 810 frames per video. This high frame count allows us to analyse the details contained in the videos more precisely and provides a significant advantage in calculating the overall video quality. The high frame count also helps us to understand how certain distortions can propagate throughout the video and allows us to assess the sensitivity of our model to types of distortion.

Another important factor that influenced our selection of the dataset is the MOS (Mean Opinion Score) scores of the videos. MOS scores are reliable scores representing the quality perceived by the human eye and allow us to validate the quality predictions produced by our model in our experiments based on real-world observations. These MOS scores provide us with the opportunity to evaluate the performance of our model on an objective basis and to analyse the compatibility of our model with human perception. The correspondence between the outputs of the model and the MOS scores is a critical indicator of the accuracy and reliability of our model, and is also very important in terms of evaluating the practical applicability of our model.

5.1.1 Data for ViNet Model

To train and evaluate the ViNet model, we used the UCF Sports Action dataset [18] and [20], which is frequently cited in the literature. This dataset was created by the Visual Computing Research Centre at the University of Central Florida and consists of 150 short video clips of athletes performing various sports actions in real-world scenarios. The dataset is available online at UCF Sports Action Dataset.

The UCF Sports Action dataset contains video clips organised into 10 different sports movement categories: diving, golf swing, kicking, weight lifting, horse riding, running, skateboarding, swing (gymnastic ring and floor swing), golf course swing and walking. Each video clip ranges in length from 1 to 10 seconds and was recorded at a standardised frame rate, which allows a detailed observation of each movement sequence. The videos were collected from television channels, making the scenes and movements close to real-world conditions 5.2.

One of the key features of this dataset is the diversity in background and mo-



Figure 5.1. MSU Video Quality Metrics Dataset

tion dynamics. Unlike controlled datasets captured in laboratory environments, the UCF Sports Action dataset contains natural and challenging environments, strengthening the model’s ability to generalise and handle complex scenes. Furthermore, since the actions are performed by multiple individuals, the dataset provides diversity in terms of body dynamics, appearance and movement styles.

Each video in the dataset is labelled with an action tag, making this data highly suitable for training supervised learning models. Thanks to this structure, models such as ViNet can learn complex movement patterns and action representations across different sports. Furthermore, the dataset contains sequences with different movement speeds, body positions and viewpoints, which is critical for the evaluation of motion recognition models.

By leveraging this dataset, the ViNet model is able to accurately recognise and predict dynamic motions, enabling robust learning and adaptability to various video data scenarios.



(a) Horse Riding Scene



(b) Swing Bench Scene



(c) Golf Scene



(d) Skateboarding Scene

Figure 5.2. UCF Sports Dataset

5.1.2 Data for FlowNet Model

To train and evaluate the FlowNet model, we used the MPI Sintel dataset [5], which is widely used in optical flow modelling studies. This dataset is based on the short animated film ‘Sintel’, which has an important place in the field of computer graphics and visual effects, and was created by the Max Planck Institute (MPI).

The MPI Sintel dataset provides an ideal resource for optical flow modelling in complex and natural environments in computer vision applications. The dataset consists of scenes with different lighting conditions, camera angles, and movement speeds. There are two basic lighting variations in this dataset: ‘Clean’ and ‘Final.’ The ‘Clean’ version provides simpler and noiseless images, while the ‘Final’ version includes images enriched with realistic effects such as shadow, fog, and motion blur, with the inclusion of environmental factors.

Every scene has rich backgrounds, moving objects, and characters that move quickly or slowly. By training on data containing a variety of movements and environmental variables, the FlowNet model is able to adjust to varied situations. Scenes with objects moving at different speeds, in particular, improve the model's ability to adapt to changing circumstances. Example of dataset can be seen in 5.3:



Figure 5.3. Sintel Dataset

The mobility of every pixel in the MPI Sintel dataset is determined by flow vectors, which are linked to each frame. The FlowNet model can learn intricate motion patterns in supervised learning procedures thanks to this labeling structure. With the help of this dataset, FlowNet gains the ability to precisely simulate the motion between each video frame, enabling it to comprehend how scenes move.

5.2 Models

As mentioned before, we use two models in our experiments: FlowNet and ViNet. FlowNet is a deep learning model for video stream prediction and ViNet is a deep learning model for video saliency prediction. For both models, pre-trained versions that have been previously trained on specific datasets are used. This approach reduces the training time and improves the performance of the

models, since both models have been trained on large datasets and have learnt basic movement and attention patterns.

5.2.1 FlowNet: Video Streaming Prediction

FlowNet is a deep learning model for optical flow estimation in computer vision. Optical flow is a technique for determining the direction and speed of motion of each pixel between two consecutive images. FlowNet is designed to solve this task with direct learning and can estimate the direction and velocity of motion with high accuracy, especially in video streaming. Optical flow estimation is used in various computer vision applications such as motion analysis, object tracking, and image segmentation.

One of the most important reasons for choosing the FlowNet model is its ability to quickly and accurately predict motion patterns in moving images. The calculation of optical flow by more traditional methods is often quite complex and time-consuming; FlowNet speeds up this process considerably, enabling practical processing of very large datasets. Since FlowNet is pre-trained on large and content-rich datasets such as the MPI Sintel dataset, it performs well in scenes with complex motion.

5.2.2 ViNet: Video Saliency Estimation

ViNet is a deep learning model developed for visual attention estimation. Visual attention estimation is based on determining the regions that attract the most attention of the viewer in an image or video. This model is specifically designed to analyse prominent objects or scenes on videos and can accurately predict the areas where attention is concentrated. The ViNet model has been pre-trained on various datasets that have been developed to understand and predict the natural viewing behaviour of humans.

We chose ViNet because of the model's ability to successfully predict areas of attention concentration, especially when working on video data. Video saliency prediction is very important in various applications such as video analysis and summarisation, attention-based video compression and user attention analyses. The use of ViNet allows us to predict the distribution of attention in video

with high accuracy. This is a critical advantage for applications requiring video analysis and attention estimation.

5.2.3 Using Different Versions Of Models

In our experiments, we preferred pretrained versions of both models. The use of pretrained models allows us to directly use generalising features that these models have previously learned on large and rich datasets. This not only reduces the training time, but also significantly improves the initial performance of the models and improves the accuracy of the results obtained in experiments. In deep learning models developed for specialised tasks, such as FlowNet and ViNet, starting with pre-trained weights enables faster adaptation to the relevant tasks and contributes to robust results.

While taking advantage of the advantages of pre-trained models, we also compared the performance of these models with scratch-trained models for future experiments. This comparison allowed us to observe how the models respond to various training strategies and to evaluate their effectiveness on specific tasks.

We used three different model variations for the FlowNet model and performed comparisons:

1. **Pre-Trained 2-Frame Model:** This is a pre-trained version of the model that is pre-trained to estimate optical flow by comparing two consecutive video frames. Since this pre-trained model for optical flow estimation was initially trained on large datasets, it has successfully learnt complex motion dynamics.
2. **2-Frame Model Trained from Scratch:** In this model, a FlowNet model that estimates optical flow by comparing two consecutive frames is trained from scratch. This resulted in a longer training time, but the results were comparable to the pre-trained model.
3. **5-Frame Model Trained from Scratch:** In order to improve the quality of the models used in our perceptual loss calculation, we designed a model trained with five consecutive frames instead of two frames. This model can recognise and predict more complex motions using a wider motion window in the video data. The model is trained from scratch and is capable of evaluating longer sequences.

For the ViNet model, we also developed and compared models trained with different frame counts:

1. **Original 32-Frame Pre-Trained Model:** This is the pre-trained version of the model that performs video saliency estimates by processing 32 consecutive frames. Trained with the original parameters, this model has the capacity to estimate attention over a wide range of frames.
2. **32-Frame Model Trained Adhering to the Original Parameters:** We created a version trained from scratch, retaining the parameters and configuration of the original ViNet model. This model was developed for comparison with the original pre-trained model and allows us to evaluate how training the model from scratch affects performance on specific datasets.
3. **4, 8 and 16-frame models trained from scratch:** In contrast to the original model, we created null-trained ViNet models with 4, 8, and 16 frames, respectively, to observe how attention changes at shorter frame intervals. The shorter frame count was used to test the model's ability to predict attention instantaneously and to observe performance on tasks that focus on shorter durations in video analysis.

The comparative experiments helped us study how well the FlowNet and ViNet models performed under different settings. We faced some computational restrictions in our calculations, which meant we had to cut down the number of batches used in the ViNet model from 8 to 4. We also reduced the number of Epochs from 40 to 10. To support our hypothesis, we trained an additional version of the models for 80 epochs. This allowed us to check the relationship between model accuracy and our scoring system. Overall, these experiments gave us a chance to look at how pre-trained models compared to models trained from scratch. The results are clearly shown in the next section.

Chapter 6

Results

The results of our study show how effective deep feature representations are for estimating video quality. We looked at the outputs from FlowNet and ViNet to see how well they could understand motion, important details, and differences in video content. We tested these models on different datasets and used metric like Spearman correlation to compare what the models predicted with what humans see.

This section explains our findings in detail. We discuss how the models performed against each other, how the length of video sequences affected accuracy, and how strong our framework is when facing different levels of distortion and noise.

Firstly, we examine the FlowNet models we trained and discuss how their performance differs from the pre-trained model. This analysis helps us evaluate the effectiveness of training and fine-tuning in improving the model’s motion estimation capabilities. Next, we analyze the correlation between the video scores derived from all FlowNet models (using the MSU dataset) and the ground truth scores. This step provides insight into how well these models align with human perception of video quality.

Following this, we repeat the same process for our saliency models, evaluating their performance and correlating their predictions with ground truth. The purpose of these analyses is to validate the hypotheses introduced at the beginning of this study. Specifically, we aim to demonstrate that adjusting video frame length and optimizing model architectures lead to improved score estimation over video quality. Through this structured evaluation, we aim to provide concrete evidence supporting our hypotheses, thereby reinforcing the validity of our ideas with empirical results.

6.0.1 Flownet Model Training and Correlation Analysis

As we can see from the figure(6.1) error calculation of flow models differs from pretrained to stretch but even after that we can clearly see in Figure 6.2, we present a scene from our experiments alongside the flow estimation for the selected scene is clearly visible and this visualization highlights the model's capability to track dynamic elements within the scene. This also proves that the model that we trained successfully demonstrates flows over scenes perfectly. Even though the performance of the model with 5 frames was worse than the other models, which only work with pairs of frames, the 5-frame model achieved a better Spearman correlation between obtained video scores and human observant scores than the 2-frame model in figure 6.3. Specifically, the weighted absolute correlation between the 5-frame model reached 0.816, outperforming the pretrained model from FlowNet, which had a weighted correlation of 0.795. Although we couldn't fully train the 5-frame flow model due to limited dataset size and computational power, even this partial training clearly shows that our second hypothesis was correct: increasing the number of input frames improves the model's correlation with ground truth.

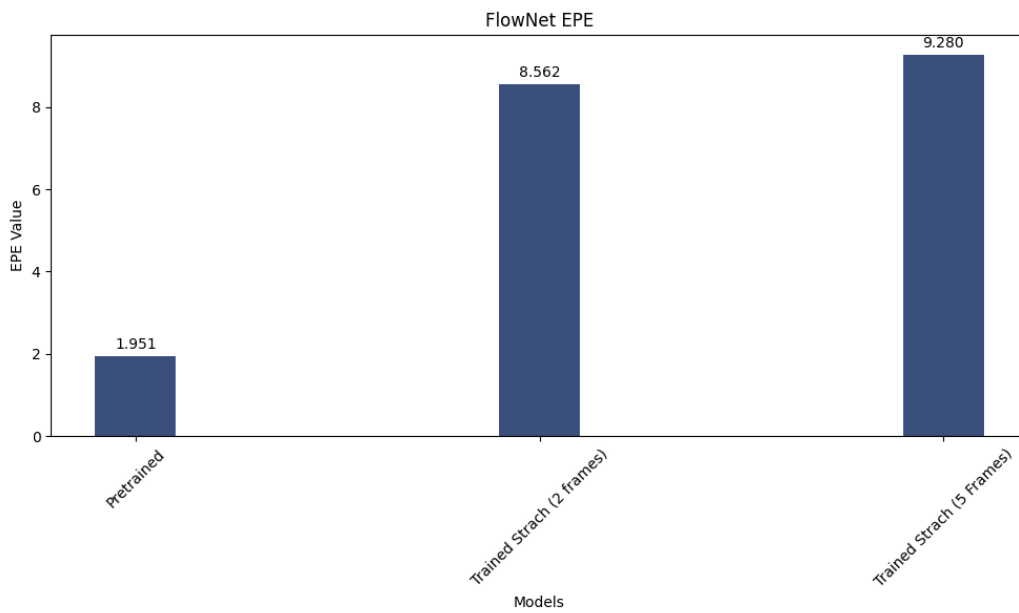


Figure 6.1. Flownet models EPE

The comparison highlights the importance of temporal length in video qual-

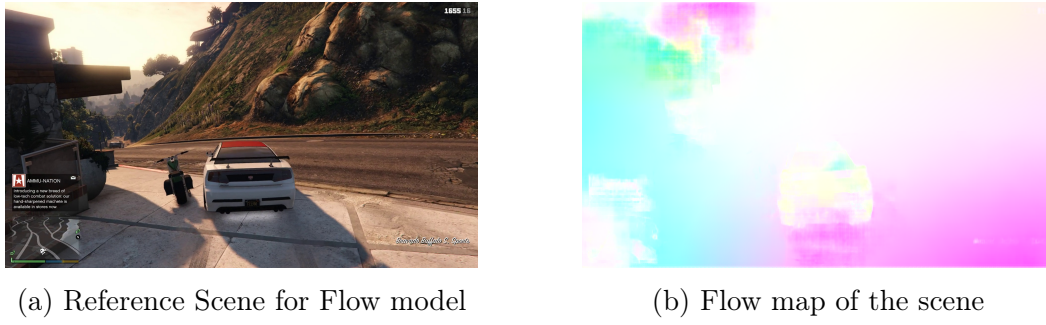


Figure 6.2. Flow example taken from experiments

ity estimation, as increasing the number of frames from 2 to 5 resulted in an improved correlation score, confirming the hypothesis that a model with more frames provides better perceptual alignment. Despite not being fully trained, the 5-frame model's performance demonstrates that additional frames can enhance model effectiveness, particularly when weighted correlation is considered. These results reinforce the hypothesis that the inclusion of more frames leads to a more accurate video quality estimation, bringing us closer to achieving human-like perception in video analysis.

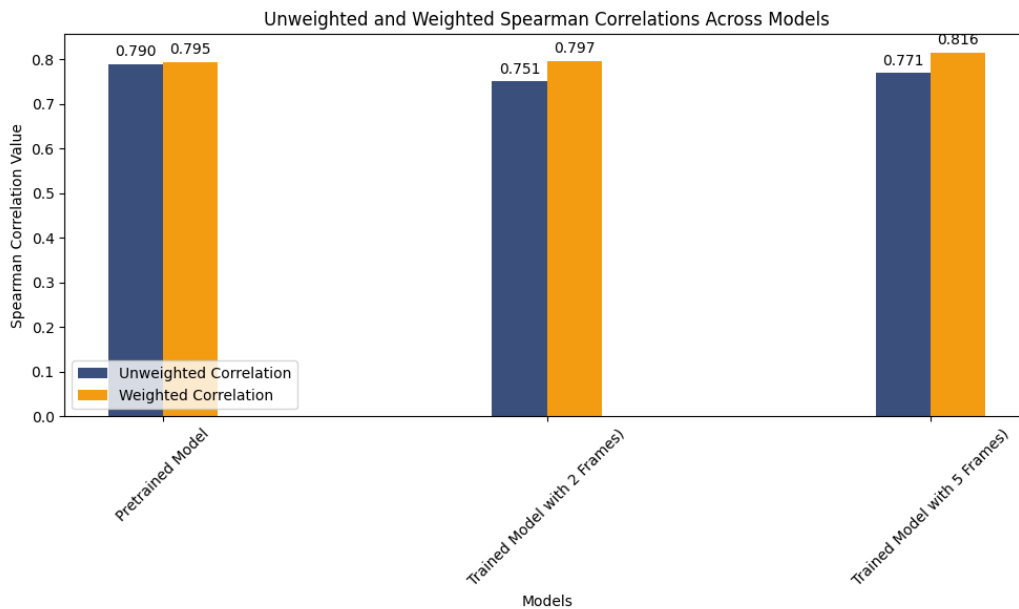
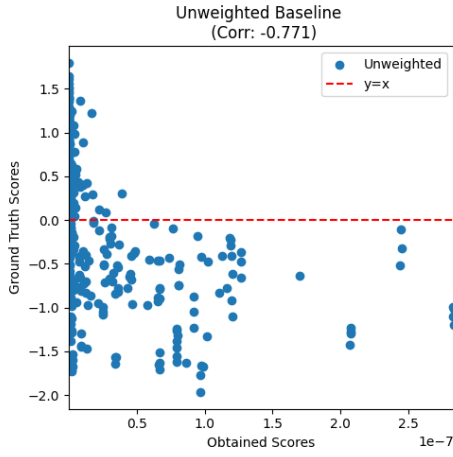


Figure 6.3. Spearman Correlation of the models

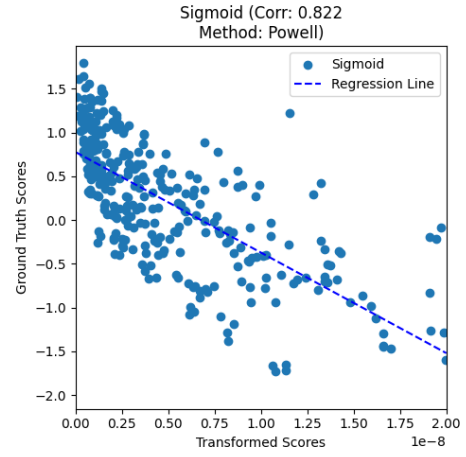
6.0.2 Flownet Model Optimization

Figure 6.4 illustrates the results of experiments conducted using different transformation functions to weight the output scores derived from the intermediate features of the FlowNet model with a 5-frame input configuration. The purpose of these experiments was to explore how various transformations could enhance the interpretability and effectiveness of the model's outputs. The unweighted distribution, depicted in Figure 6.4a, served as a baseline for comparison, showing the raw scores without any adjustments. Among the transformation functions tested, the sigmoid function (Figure 6.4b) demonstrated superior performance due to its ability to normalize values into a smooth, bounded range while emphasizing critical differences near the extremes. Other functions, such as the softmax (Figure 6.4c) and arc tangent (Figure 6.4d), offered alternative approaches to scaling, but they were less effective in achieving consistent results.

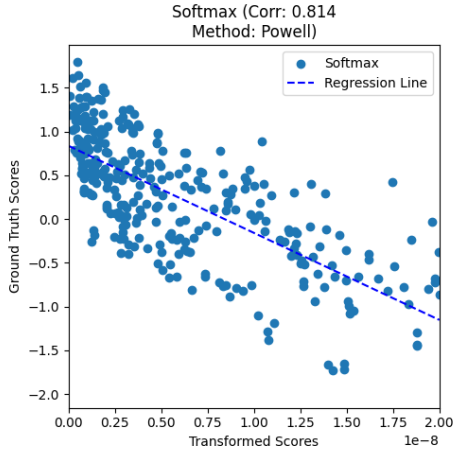
The exponential transformation (Figure 6.4e) amplified score differences but introduced instability, while the linear transformation (Figure 6.4f) preserved proportional relationships but lacked the adaptability needed for optimal performance. To further refine the results, several optimization algorithms were applied, including Powell, L-BFGS-B, TNC, Nelder-Mead, and SLSQP. Among these, the Powell method emerged as the most effective due to its robustness and ability to optimize the transformed scores efficiently. Combining the sigmoid transformation with Powell optimization consistently yielded the best results, significantly improving the correlation of model outputs with perceptual quality metrics. These findings highlight the importance of choosing appropriate transformations and optimization techniques to enhance the alignment of computational predictions with human visual perception, offering a meaningful step forward in perceptual quality estimation.



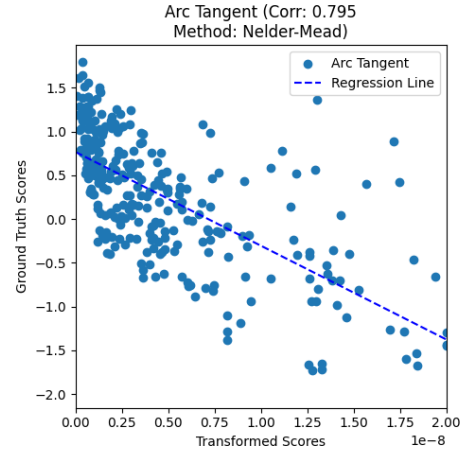
(a) Unweighted distribution of scores



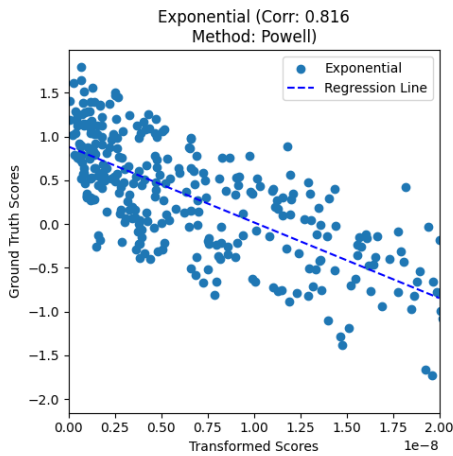
(b) Sigmoid fitting



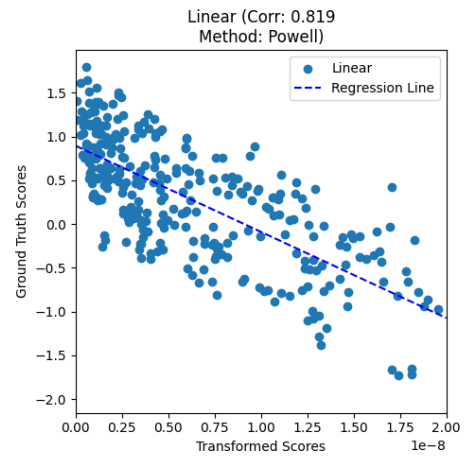
(c) Softmax fitting



(d) Arc tangent fitting



(e) Exponential fitting



(f) Linear fitting

Figure 6.4. Flownet model output figures from 5 frame input model

6.0.3 Saliency Model Training and Analysis

Figure 6.5 shows the similarity metrics and coefficient correlation scores for the ViNet model after 10 training epochs. Figure 6.6 presents similar results for the model trained for 80 epochs. These figures highlight important patterns in how the ViNet model performs with different input frames, specifically 4, 8, and 16 frames. As we can see the best-performing model was again a pretrained 32-frame model after that other best-performing one was 16 frames this also defends our theory about frame length positively affects model performance.

In the experiment with 10 epochs on the figure 6.7 we can clearly observe that the 8-frame setup did better than the 16-frame setup. This is surprising since more frames are usually expected to provide better context over time. This unusual result likely comes from having a small dataset and limited computing power. Longer sequences need a bigger and more varied dataset to properly understand the timing between frames and enough resources to train effectively. When these resources are not available, the model has a hard time generalizing with longer inputs, resulting in lower correlation scores. This shows that it is important to balance the length of the input with the available data and computing capabilities.

The results of figure 6.8 also back up a second idea: models that do well by evaluation metrics tend to get higher perception correlation scores. For example, as training increased from 10 to 80 epochs, all input configurations (4, 8, and 16 frames) showed small drops in their unweighted Spearman correlation scores. For 4-frame inputs, the score decreased from 0.817 to 0.796; for 8-frame inputs, from 0.825 to 0.815; and for 16-frame inputs, from 0.817 to 0.80. This decline suggests that training the model too much can harm its ability to match human judgments. A model that cannot understand motion or important details better will have a worse perceptual correlation, indicated by lower scores.

These results show the important choices in video saliency modelling. They stress the need for a large dataset, enough computational power, and the right amount of temporal length to reach the best performance.

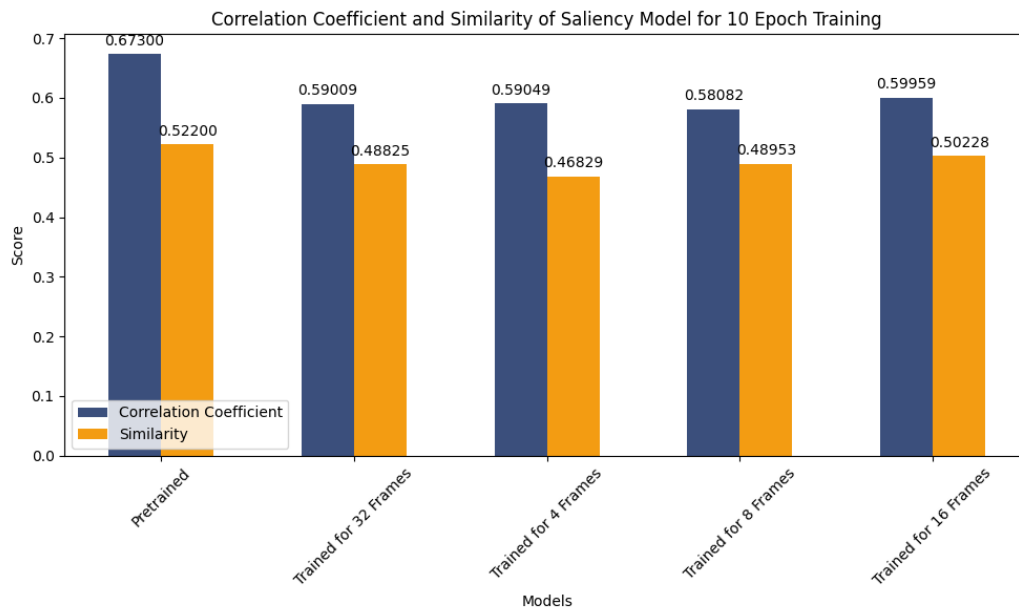


Figure 6.5. Scores for 10 Epoch Trained Model

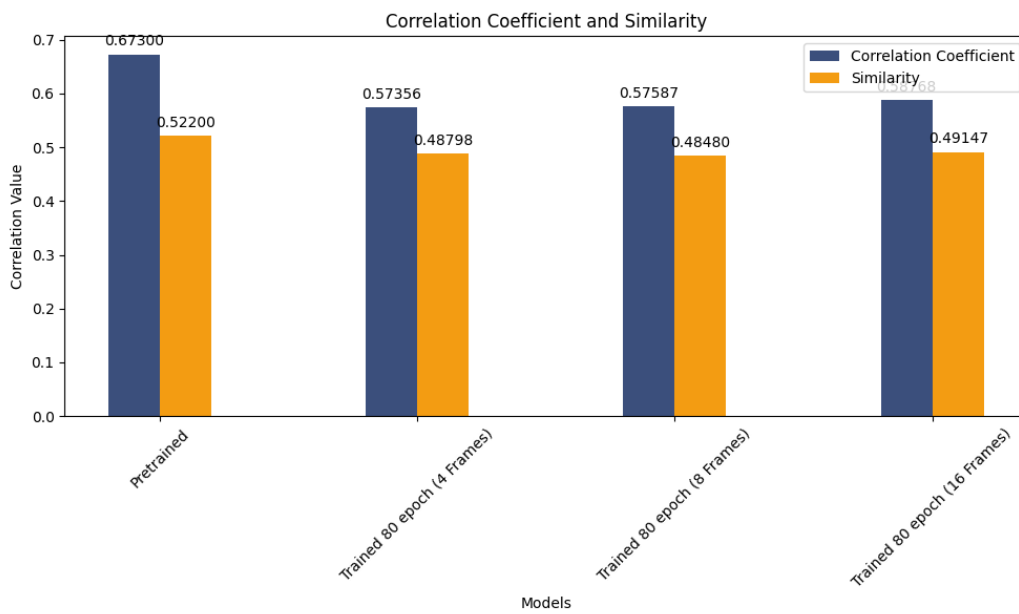


Figure 6.6. Scores for 80 Epoch Trained Model

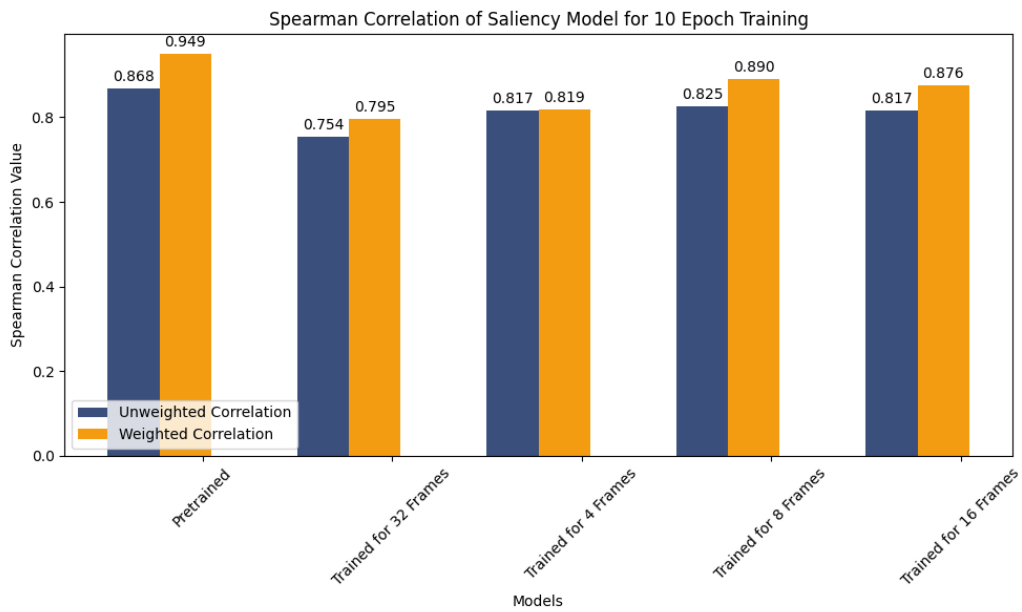


Figure 6.7. Spearman Correlation for 10 Epoch Trained Model

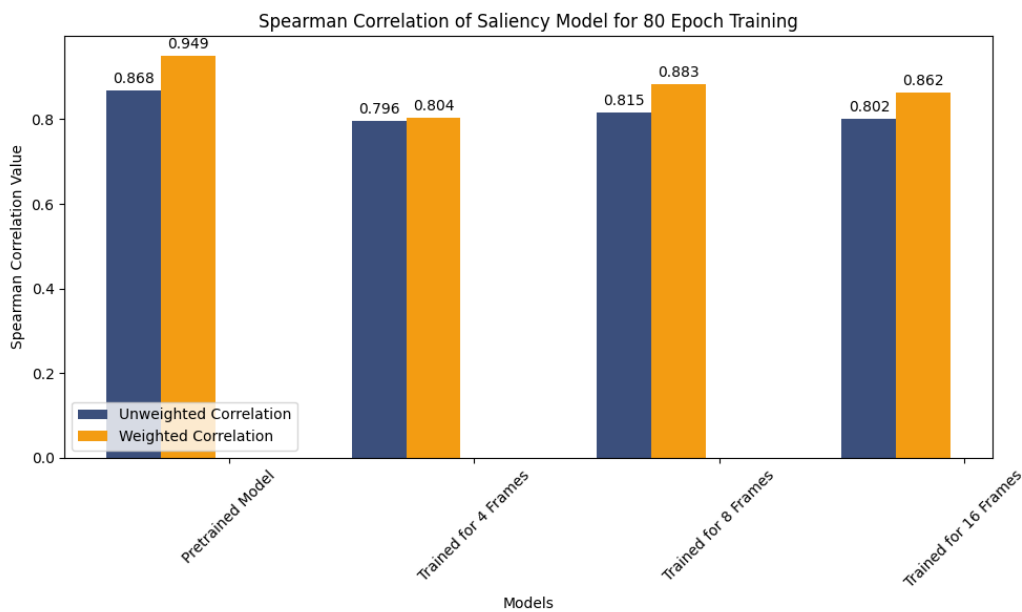


Figure 6.8. Spearman Correlation for 80 epoch Trained Model

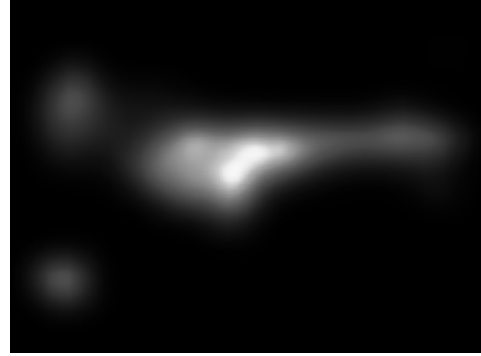
We can also visually compare the saliency models with input frames, as demonstrated in the example shown in Figure 6.9. However, this process often

requires additional steps, such as deblurring or other preprocessing techniques, which are beyond the scope of this study. Therefore, we include it here purely as a demonstrative example.

Even without further processing, the results clearly highlight significant features in the scene. For instance, the plane's outer lines and the character are more prominent compared to the surrounding areas, indicating the saliency model's ability to focus on meaningful regions effectively.



(a) Reference Scene for Saliency model



(b) Saliency map of the scene

Figure 6.9. Saliency Map example taken from experiments

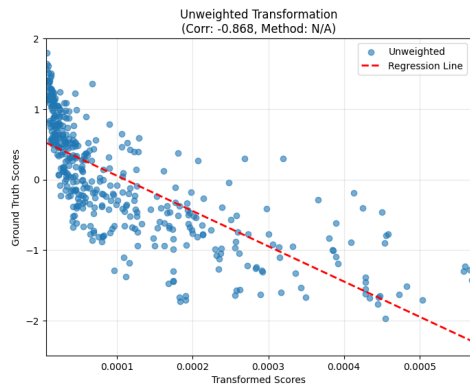
6.0.4 Saliency Model Optimization

The necessity of non-linear fits arises from the inherent nature of the prediction task. The predictions generated by our models do not need to exhibit a direct linear correlation with human judgments. Therefore, it is crucial to account for these non-linear relationships by employing psychometric functions, which are specifically designed to model perceptual phenomena.

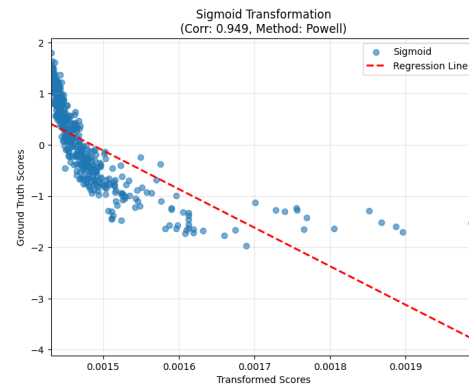
After analyzing the FlowNet model, we encountered a similar challenge with the ViNet model, as illustrated in Figure 6.10. Due to the complexity of the results and the extensive number of experiments conducted, it was impractical to include every output figure in this thesis. For both models, we tested various transformation functions and optimization algorithms to determine the most effective configuration for aligning model outputs with perceptual metrics.

Through extensive experimentation, we identified the sigmoid transformation function and the Powell optimization method as the most effective combination for the ViNet model. This configuration ensured a robust alignment between the model predictions and perceptual quality metrics. Given the high

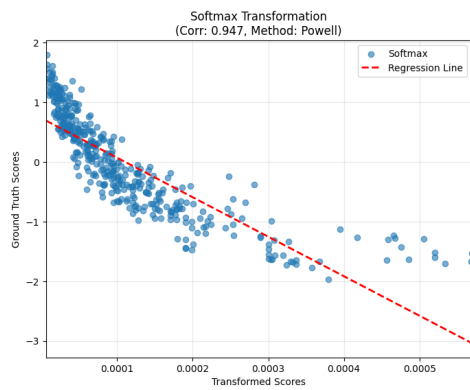
volume of experiments, we have chosen to summarize the key results rather than display individual output figures for every tested configuration. This approach highlights the critical role of selecting the appropriate transformation and optimization methods to achieve strong correlations with perceptual quality metrics. Furthermore, it mirrors the methodology employed for FlowNet, reinforcing the broader applicability of the proposed techniques across different models.



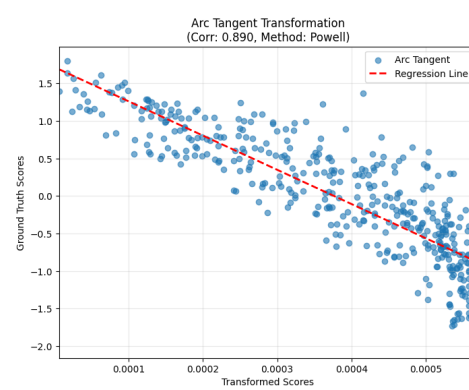
(a) Unweighted distribution of scores



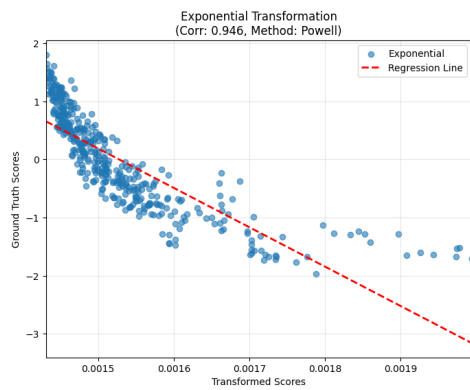
(b) Sigmoid fitting



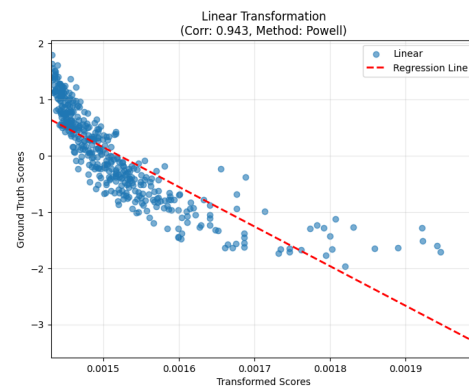
(c) Softmax fitting



(d) Arc tangent fitting



(e) Exponential fitting



(f) Linear fitting

Figure 6.10. Vinet model output figures from pretrained model

6.0.5 Approach Submission

Our approach was tested using the MSU Media dataset [2], a global full-reference benchmark for video quality metrics. For evaluation, we submitted our pre-trained saliency model and Flow model that we trained from scratch with 5 frame input, Saliency which processes sequences of 32 frames as input to compute saliency maps. Those specific models was chosen because it consistently outperformed other variations in our experiments. The extended sequence length allows it to capture temporal dynamics more effectively, resulting in significantly improved accuracy in estimating video quality compared to models using shorter frame sequences. The Saliency model scored a Spearman correlation of 0.949. This shows a strong match with human scores. It did better than full-reference metrics like YUV-VMAF, which scored 0.9423. It also outperformed no-reference methods like NR MDTVSFA, which had a score of 0.929. In comparison to the deep learning models from previous studies, which aimed at improving quality but needed more computing power, this method provides similar or better performance. It also uses less computational power while keeping good perceptual quality. Our method performs well in real-world situations. It aligns with human evaluations, showing its effectiveness in both full-reference and no-reference situations. Important points from the MSU Media dataset include its ability to adapt to various uses. The primary reason our method outperforms others lies in its design to effectively utilize deep features and saliency-driven features. By incorporating longer sequences of frames, our model captures richer spatiotemporal information, improving its sensitivity to distortions and motion inconsistencies. Unlike many traditional methods that rely on pixel-level or limited-frame analysis, our framework integrates advanced deep learning models FlowNet and ViNet to leverage their ability to extract and process dynamic, high-level features. This helps it overcome the challenges that traditional compression methods, like H.264 and HEVC, face in maintaining quality while also being efficient. As illustrated in the figure below (Figure 6.11) and detailed on the official website of the benchmark, our approach was submitted for evaluation, providing a comprehensive comparison with existing methods in the field.

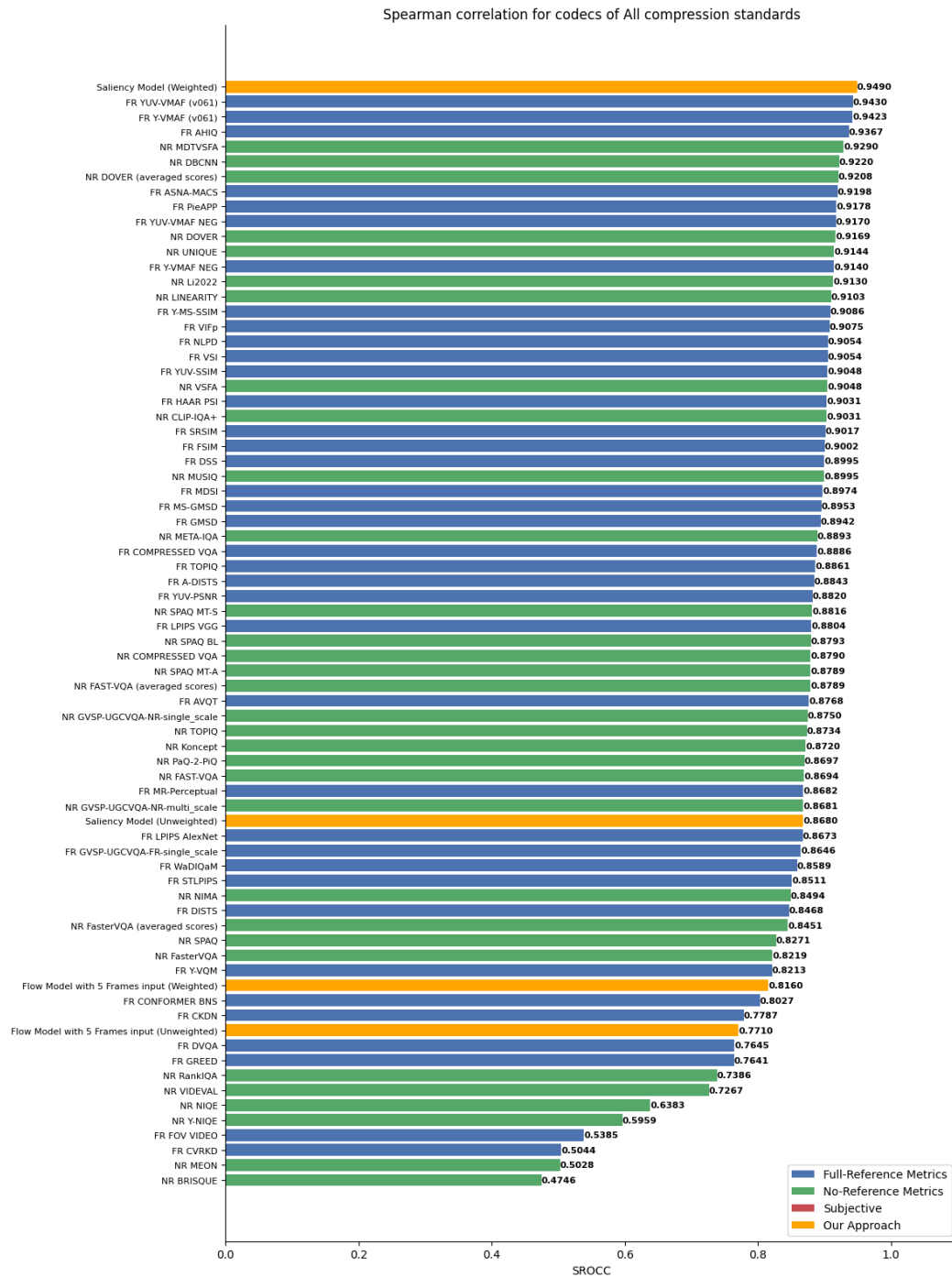


Figure 6.11. Model's Performance comparing publicly released Compression standards of MSU Graphics&Media lab competition [1]

Chapter 7

Discussion

Our approach revealed both strengths and limitations when we looked at how FlowNet and ViNet performed in measuring video quality. By using features that consider time and areas that catch the viewer’s attention, we found a strong link between what computers measure and what humans see. This connection is important for making video quality assessments more focused on human experience. Still, we faced challenges, especially with extreme distortions and the need for better computational efficiency. These challenges show us where we can improve.

A big achievement in this study was using temporal length to grasp video dynamics. By examining a series of frames, the models managed to maintain motion coherence and pinpoint important regions across frames. This is vital for how humans perceive video content. ViNet successfully highlighted attention-driven areas, showing that understanding space is key to perceptual quality. At the same time, FlowNet’s optical flow estimates gave us a dependable idea of motion accuracy, spotting distortions that traditional methods often miss. By combining these models, we could assess video quality in a way that considers both space and time. This approach is essential for connecting computer assessments with how people actually experience videos.

To validate the framework, we designed experiments that leveraged feature maps extracted from both clean and corrupted video frames. A pixel-wise comparison of these feature maps quantified perceptual loss, providing a novel approach to assess distortions such as compression artifacts and noise. Beyond simply correlating these metrics with human-perceived scores, we optimized the alignment by weighting the feature-based differences with transformation functions such as sigmoid, softmax, and arc tangent. These transformations were selected for their ability to normalize and scale feature differences in a manner

that better reflects human perceptual sensitivity. This optimization step was instrumental in bridging the gap between algorithmic evaluations and human judgment, ensuring that our metrics were both accurate and interpretable.

Despite these successes, several limitations emerged. Extracting and analyzing intermediate feature maps posed significant computational challenges, limiting the framework’s applicability in real-time scenarios. This computational overhead becomes especially problematic in contexts requiring immediate feedback, such as video streaming or live video enhancement. Additionally, the reliance on benchmark datasets, while offering controlled evaluation environments, introduced constraints in generalizability. The datasets employed lacked the dynamic range and complexity seen in real-world applications, underscoring the need for future studies to incorporate more diverse and challenging content. The original pre-trained saliency model, which was trained for over 40 epochs, did better than our results. This happened because it had a longer training period and a larger dataset. However, in our experiments, the setup with 8-frame input worked the best. This shows that having a good amount of time in the video helps improve the results, as long as there is enough data and training resources available. We could not test longer inputs, like 32 frames, because of time and computing limits.

The models encountered challenges when handling videos with extreme distortions, such as significant noise or heavy compression artifacts. These circumstances occasionally resulted in differences between model predictions and human perceptual assessments, indicating that current feature extraction methods may not adequately capture all subtleties of perceptual quality. Additionally, the lack of targeted optimization for real-world distortions in the training datasets restricted the models’ adaptability in such difficult conditions.

Future investigations could concentrate on utilizing advanced video techniques such as super-resolution models. These models have the potential to enhance spatial resolution, resulting in detailed images of larger structures. This enhancement is crucial for high-quality video content. Combining deep learning-based metrics with traditional perceptual quality models may yield a more balanced approach. Such a blend would be particularly beneficial for managing significant distortions.

This research demonstrates that deep features in pretrained networks, particularly for tasks involving the time domain, effectively distill temporal features critical for perception. These features, extracted from intermediate layers, encapsulate essential spatiotemporal dynamics that correlate strongly with video quality. While previous studies established the utility of such features for images, this work extends that understanding to videos, marking a significant ad-

vancement in the field.

The findings underscore the ability of deep feature maps to align computational evaluations with human perception by capturing temporal coherence and motion-based nuances. To fully exploit this approach, challenges related to computation and datasets must be addressed. Expanding datasets to include a broader range of dynamic video content will improve the robustness of these methods. Furthermore, enhancing computational efficiency will facilitate their deployment in practical applications such as adaptive streaming, video compression, and real-time video enhancement.

Chapter 8

Conclusion

In this thesis, we delved into the integration of deep learning models to advance the field of perceptual video quality estimation. Leveraging the strengths of FlowNet for optical flow computation and ViNet for saliency prediction, we developed a comprehensive framework that bridges computational metrics with human visual perception. By focusing on deep feature representations, our approach highlights the critical role of spatial and temporal dimensions in assessing video quality, addressing key limitations in traditional pixel-based methods.

Our findings demonstrate that incorporating motion coherence and saliency-driven features into the analysis can significantly enhance the alignment between computational evaluations and human perception. This is particularly evident in how distortions such as compression artifacts, noise, and motion inconsistencies are captured and analyzed. The proposed methodology not only improves the accuracy of quality assessment but also emphasizes the importance of understanding the nuanced interactions between frame-level details and temporal dynamics in videos.

Beyond the technical contributions, this work has practical implications for industries reliant on high-quality video processing, including streaming platforms, real-time video enhancement systems, and content delivery networks. By providing a framework that links model predictions with viewer experiences, our research offers a pathway for improving video delivery standards and enhancing user satisfaction.

While the outcomes of this study are promising, they also open avenues for future research. Further optimization of feature extraction techniques and exploration of more advanced model architectures could provide even greater accuracy and efficiency in perceptual quality estimation. Additionally, extending the framework to encompass diverse video content, including high-motion

sequences and varying resolution levels, would broaden its applicability. Investigating how these methods interact with emerging technologies such as virtual reality, augmented reality, and immersive media could further push the boundaries of perceptual video quality analysis.

In conclusion, this thesis underscores the transformative potential of deep learning models in video quality estimation. By aligning computational techniques with the complexities of human perception, it offers a robust foundation for both academic inquiry and practical advancements. As the demand for superior video experiences continues to grow, our contributions showed the way for innovative solutions that enhance both the technical precision and perceptual fidelity of video content. This work marks a step forward in the journey toward enriching video experiences across diverse applications, setting the stage for a new era of video quality analysis and enhancement.

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